

Highlights

Overseas Remittances and Resilience to Adverse Shocks: Causal Evidence from Smallholder Households in Nepal

- A district-level Bartik shift-share IV isolates the causal effect of remittances.
- Overseas remittances significantly reduce food insecurity: HFIAS decreases 0.825, FIES 0.610 pts.
- IV-TWFE shows remittances buffer smallholders from shock-driven welfare losses.
- A kernel-weighted PSM-DiD confirms insurance effects independent of the Bartik IV.
- Insurance effects are concentrated among smallholder households and operate through consumption smoothing rather than dietary upgrading.

Overseas Remittances and Resilience to Adverse Shocks: Causal Evidence from Smallholder Households in Nepal

Abstract

This paper estimates the causal effects of overseas remittances on household welfare and their capacity to buffer smallholder households against shock-induced losses in Nepal. Using the Nepal Household Risk and Vulnerability Survey (HRVS) panel 2016–2018 and a Bartik shift-share instrumental variable built from pre-2016 district-level migration destination shares interacted with destination-country GDP growth, we address endogeneity in remittance receipt and amount.

Across ten welfare outcomes, IV-2SLS estimates show that a one-log-unit increase in overseas remittances significantly reduces food insecurity, lowering HFIAS by 0.825 points ($p < 0.001$), FIES by 0.610 points ($p < 0.001$), and moderate-to-severe food insecurity by 10.6 percentage points ($p < 0.001$), while increasing the Food Consumption Score by 6.41 points ($p = 0.063$) and log per capita food value by 0.098 ($p = 0.047$). An IV-TWFE interaction model shows that remittances buffer shock-induced welfare losses: the buffering coefficient θ is -1.045 ($p = 0.005$) for HFIAS and -0.753 ($p = 0.001$) for FIES, significantly offsetting the shock-induced increase in reduced consumption ($\theta = -0.118$, $p = 0.031$). The estimates are robust to alternative remittance measures and a kernel-weighted PSM-DiD estimator, which confirms consumption smoothing (ATT = 0.042, $p = 0.006$) and insurance effects (ATT = -0.018 , $p = 0.084$) independent of the Bartik instrument.

These insurance effects concentrate among smallholder households, which face identical shock prevalence but greater food insecurity than non-smallholders, operating through consumption smoothing rather than dietary upgrading. The findings establish overseas remittances as an informal insurance mechanism, with implications for remittance-market efficiency, labor-constrained agricultural adjustment, and shock-responsive social protection design in remittance-dependent economies.

Keywords: Overseas remittances, Food security, Smallholder households, Instrumental variable, NELM, Nepal

JEL: F24, I32, O12, Q12

1. Introduction

International remittances are now a major force in global development. In 2022, global remittance flows reached USD 831 billion, nearly three times the volume of official development aid (IOM, 2024). Nepal received approximately 11 billion dollars in overseas remittances in 2023, equivalent to 26.6% of its gross domestic product (World Bank, 2024).

The country ranks ninth globally and first in South Asia in terms of remittances as a share of GDP, highlighting the strong dependence of household incomes and national welfare on migration income (Ratha, 2023; World Bank, 2024). In a small, landlocked, and shock-prone economy such as Nepal, these transfers constitute a structural pillar of the macroeconomy and of household welfare, rather than a supplementary income source. Remittances have also contributed substantially to poverty reduction in Nepal during the 2011–2023 period by preventing approximately 2.6 million people from falling into poverty (World Bank, 2024). Understanding the welfare effects of remittances is therefore a key policy priority, particularly for developing countries such as Nepal.

Nepal’s rural smallholder households are at the intersection of two distinct risk dimensions. On the income side, more than one-fifth of the national population (20.27%) remains below the national poverty line (Central Bureau of Statistics (CBS), 2024), and a large share of the near-poor are marginal farmers whose livelihoods depend mainly on landholdings of half a hectare or less. On the shock side, more than 44% of rural households report experiencing an adverse shock in a given year, including droughts, floods, price increases, illness, and death in the family (De Stefani et al., 2022). Smallholders are distinguished from larger landholders not by their frequency of shock exposure but by their absorptive capacity. With an average productive area of 0.30 hectares, binding credit constraints, and negligible access to formal insurance, smallholder households have almost no buffer against welfare deterioration when a shock occurs (Mainali and Tuladhar, 2025). This underlying vulnerability, rather than greater shock exposure, makes the insurance role of remittances central. For households that receive them, overseas remittances can serve as an important informal safety net.

The theoretical foundation for studying remittances as an informal insurance mechanism is provided by the New Economics of Labor Migration (NELM), originally developed by Lucas and Stark (1985) and Stark and Bloom (1985), and later synthesized by Taylor (1999), who re-conceptualized migration not as an individual income-maximizing decision but as a household strategy for managing risk. Within this framework, households diversify income sources by sending a member to work in a geographically separate labor market, where earnings are less exposed to local economic conditions. The NELM also predicts a countervailing channel: migration reduces on-farm labor supply, which can depress agricultural output and food availability before remittances accumulate enough to compensate. The sign and magnitude of the net welfare impact of remittances are therefore theoretically ambiguous and cannot be resolved without credible causal identification that separates the positive income effect from the negative labor-withdrawal effect.

Remittances function as informal insurance, with counter-cyclical flows well-documented across contexts. Yang and Choi (2007) show that rainfall-induced income shocks in the Philippines triggered compensatory remittance increases, consistent with the predictions of the NELM, while Combes and Ebeke (2011) find that remittances significantly dampen the instability of household consumption in 87 developing countries. A study by Deyal (2024a) further documents systematic surges of remittances following natural disasters. The robustness of this counter-cyclical pattern was further underscored during the COVID-19 pandemic, when remittance flows decreased by only 1.1% despite a 3% contraction in global

income (Ratha, 2023), and remittance-receiving households maintained significantly better income positions than their non-receiving counterparts (Abidi et al., 2023).

Household-level welfare effects are comparably well-documented, although the strength of causal identification varies substantially across studies. A 10% increase in per capita remittances reduces poverty headcount by 3.5% in 71 developing countries (Adams Jr and Page, 2005), with magnitudes comparable to those of more recent data (Azizi, 2019). Among the welfare dimensions, food security is where the evidence is most compelling. Mora-Rivera and van Gameren (2021) use IV ordered probit models to establish that remittances causally reduce HFIAS scores among rural households in Mexico, while Alam et al. (2025) find that the receipt of remittances reduces food insecurity by more than twofold in Bangladesh using doubly robust methods. Positive effects on food expenditure and post-shock consumption are similarly documented in Sri Lanka (Jayaweera and Verma, 2024) and Nigeria (Otame, 2023).

Although Nepal-specific evidence is broadly consistent with these cross-regional findings, it remains limited in rigor of identification. Rural households receiving remittances are 2.3% less likely to fall into poverty (Nepal Rastra Bank, 2021); remittances contributed significantly to the reduction in poverty during 1996-2011 (Wagle and Devkota, 2018), and are positively associated with spending on food, education, and healthcare (Thapa and Acharya, 2017). The most recent study using the same HRVS 2016–2018 panel finds a significant increase driven by remittances in food and non-food expenditure through propensity score matching (Thapa-Parajuli et al., 2025), but does not address the endogeneity of remittance amounts, does not condition welfare effects on shock exposure, and does not test whether remittances buffer households against shock-induced welfare losses, which is the central prediction of the NELM insurance hypothesis.

The causal interpretation of the existing findings for Nepal is constrained by two compounding identification challenges. The first identification challenge stems from the non-random nature of remittance receipt. Households that send a member abroad are not a representative cross-section of the population; they tend to be systematically wealthier, better educated, and more socially connected at baseline, and these same characteristics would independently improve household welfare irrespective of migration status (McKenzie and Rapoport, 2010; Alam et al., 2025). This positive selection drives the OLS estimates upward, making remittances look more beneficial than they actually are. The second challenge cuts in the opposite direction. The NELM framework points to a labor-withdrawal effect: when a working-age household member leaves for overseas work, the household loses a pair of hands on the farm. On-farm labor supply contracts, and agricultural output can suffer before remittance income arrives in large enough amounts to make up the difference. This trade-off has been documented directly among Nepalese smallholder wheat producers, where migration increases remittance income but significantly reduces wheat productivity due to labor shortages (Paudel et al., 2025), and a similar opposing dynamic has been identified in Bangladesh (Romano et al., 2020). Because these forces operate in opposing directions, OLS estimates of the remittance–welfare relationship carry an indeterminate net bias, depending on which force dominates, they might understate, overstate, or even reverse the sign of the true effect (Fitriani and van Rijn, 2026). Existing welfare conclusions based on

such estimates therefore provide an unreliable basis for policy design. Most existing Nepal studies are based on cross-sectional data (Thapa and Acharya, 2017; Nepal Rastra Bank, 2021), propensity score matching (Thapa-Parajuli et al., 2025), or pseudo-panel methods (Acharya and Leon-Gonzalez, 2012; Wagle and Devkota, 2018), none of which resolves these challenges. No existing study provides a causal test of whether remittances buffer Nepalese smallholder households against shock-induced welfare losses—a gap especially consequential where remittances constitute 26.6% of GDP and more than 44% of rural households report an adverse shock annually. This paper therefore asks: do overseas remittances causally reduce food insecurity among smallholder households in Nepal, and do they buffer these households from shock-induced welfare losses?

This paper addresses these challenges by estimating both the causal welfare effects of overseas remittances and their role as informal insurance against adverse shocks. The analysis uses the Nepal Household Risk and Vulnerability Survey (HRVS) 2016–2018, a nationally representative three-wave household panel designed to capture shock exposures, coping strategies, and welfare outcomes (Walker et al., 2019). To address endogeneity, we construct a shift-share Bartik instrumental variable following the methodological frameworks developed by Goldsmith-Pinkham, Sorkin, and Swift (2020) and extended by Borusyak, Hull, and Jaravel (2022). The instrument interacts with pre-2016 district-level migration destination shares with lagged annual real GDP growth rates in destination countries.

This identification strategy exploits plausibly exogenous variation in destination-country economic conditions to estimate the causal effect of remittances on household welfare. Nepal’s minimal economic integration with its main migrant-destination countries in trade, foreign direct investment, development aid, and tourism (World Bank, 2025; Department of Industry, 2025; Nepal Economic Forum, 2025) ensures that destination-country GDP growth shocks reach Nepalese households through no meaningful pathway other than remittances, after conditioning on fixed effects in the household and year. Instrument strength and endogeneity diagnostics support the use of the IV approach. The paper then estimates an instrumental-variable two-way fixed effects model to test whether remittances provide insurance against shocks. In this model, both remittances and the remittance–shock interaction are treated as endogenous. They are instrumented using the Bartik instrument and its interaction with the shock indicator, respectively, following the standard two-instrument approach for endogenous interaction terms (Heckman and Vytlacil, 2005; Wooldridge, 2010; Bun and Harrison, 2019). The coefficient of the remittance–shock interaction captures the buffering effect of remittances and is the key parameter of interest. As an additional robustness check, the study also implements a kernel-weighted propensity score matched difference-in-differences estimator, which does not rely on the Bartik instrument.

The analytical focus on smallholder households, defined here as those cultivating 0.5 hectares or less, comprising 71% of the sample, reflects a deliberate theoretical choice that the data validate. Smallholders in Nepal face food insecurity scores (HFIAS) nearly twice as high as non-smallholder households (1.073 versus 0.553) and are 5.8 percentage points more likely to experience moderate-or-severe food insecurity, despite facing statistically identical aggregate shock prevalence (44.6 versus 43.8%, $p = 0.317$) and identical remittance receipt rates (57.4 versus 58.2%, $p = 0.439$). The welfare gap between the two groups therefore

reflects differential coping capacity rather than differential risk exposure. This structural logic implies that overseas remittances, for smallholder households specifically, are not supplementary income but the primary buffer against welfare deterioration, and that the causal question about insurance effects is most consequential, and most identifiable, within this stratum.

This paper makes four contributions to the empirical literature on remittances and household welfare. First, it provides the first quasi-experimental panel evidence on the causal welfare effects of overseas remittances in Nepal, using a Bartik shift-share IV with strong first-stage diagnostics to address positive selection and reverse causality. Second, it provides the first household-level causal test of the NELM insurance hypothesis in Nepal, and to our knowledge, in any South Asian smallholder context, by estimating the shock-buffering effect of remittances using an IV two-way fixed effects model with two endogenous regressors. Third, the analysis evaluates ten welfare outcomes spanning four conceptually distinct dimensions, namely food consumption value, dietary quality, dietary diversity, and experienced food insecurity, identifying which welfare channels remittances protect under shock and which they do not. Fourth, it distinguishes the extensive margin of remittance receipt from the intensive margin of the remittance amount, exploiting the theoretical prediction that labor-withdrawal costs dominate at the extensive margin while income accumulation effects emerge at the intensive margin—a distinction that helps reconcile conflicting findings in the prior literature.

The remainder of the paper is organized as follows. Section 2 describes the data source, the construction of the variables and empirical strategy, including the construction of the Bartik instrumental variable. Section 3 presents the empirical results, beginning with descriptive evidence on smallholder vulnerability, followed by the IV-2SLS direct effects and the IV-TWFE buffering model. The section also reports robustness checks using a kernel-weighted PSM-DiD estimator. Section 4 discusses the findings in relation to the framework of the New Economics of Labor Migration and the broader literature. Section 5 concludes with policy implications.

2. Materials and Methods

2.1. Data Source and Sample

This study uses data from the Nepal Household Risk and Vulnerability Survey (HRVS) 2016–2018, a nationally representative three-wave panel dataset, collected by the World Bank in collaboration with the Government of Nepal ([Walker et al., 2019](#)). The survey covers approximately 6,000 households in 400 communities (Primary Sampling Units, PSUs) drawn from 50 of 75 districts using probability-proportional-to-size (PPS) sampling, explicitly excluding Kathmandu Valley and the Mountain ecological zones to focus on rural and semi-urban households most exposed to livelihood shocks. The three waves (2016, 2017, 2018) produce a balanced panel with a three wave retention rate of 94% and a wave 2 retention rate of 97%, producing 18,537 household-year observations across 6,367 unique households. The analytical sample is restricted to marginal or very small smallholder households (≤ 0.5 ha), a lower-bound threshold widely used to define small farmers in the South

Asian agricultural literature ([Gc and Hall, 2020](#); [Prins et al., 2022](#)). This subsample comprises 71% of all observations ($N = 13,169$ household-wave observations from 4,619 unique households) and represents the most vulnerable, migration-dependent segment of Nepal’s agricultural population. In addition, analysis is conducted for the non-smallholder subsample (> 0.5 ha) and the entire pooled sample to assess heterogeneity of the treatment effect across landholding strata.

2.2. Welfare Outcome Variables

To ensure a comprehensive assessment of the welfare effects of remittances, this study examines ten welfare outcomes spanning four conceptually distinct dimensions: food consumption value, dietary quality, dietary diversity, and experienced food insecurity. This multi-dimensional approach reflects the recognition that welfare in the context of income shocks cannot be captured by any single indicator, and that different dimensions may respond differentially to remittance flows ([Jones et al., 2013](#); [Meliko et al., 2023](#)). The value of food consumption is measured by (i) the log per capita weekly food value and (ii) the log total annual food expenditure. Both capture household food acquisition from all sources (purchased, home-produced, in-kind) and are log-transformed to reduce skewness and yield proportional-change interpretations. Dietary quality is assessed by the Food Consumption Score (FCS), a WFP composite indicator weighted by nutritional importance in eight food groups ([WFP, 2008](#)):

$$\text{FCS}_{it} = 2 \cdot \text{Staples} + 3 \cdot \text{Pulses} + 1 \cdot \text{Veg} + 1 \cdot \text{Fruit} + 4 \cdot \text{Protein} + 4 \cdot \text{Dairy} + 0.5 \cdot \text{Sugar} + 0.5 \cdot \text{Oil} \quad (1)$$

Where each term represents the frequency of consumption (days, 0–7) in the preceding week. The Per capita FCS is calculated by dividing the household FCS by household size.

Dietary diversity is measured by the Household Dietary Diversity Score (HDDS), which counts the number of distinct food groups consumed in the preceding 24 hours across 12 standardized FAO food groups ([Kennedy et al., 2011](#)):

$$\text{HDDS}_{it} = \sum_{g=1}^{12} \delta_{igt}, \quad \delta_{igt} \in \{0, 1\} \quad (2)$$

Experienced food insecurity is captured by two validated instruments. The Household Food Insecurity Access Scale (HFIAS) sums nine frequency-weighted items (scored 0–3 on a “never–often” scale) measuring anxiety, quality reduction, quantity reduction, and deprivation ([Coates et al., 2007](#)):

$$\text{HFIAS}_{it} = \sum_{j=1}^9 \text{freq}_{ijt}, \quad \text{freq}_{ijt} \in \{0, 1, 2, 3\} \quad (3)$$

The Food Insecurity Experience Scale (FIES) sums eight binary items (yes/no) ordered from the least severe to the most severe manifestations of food insecurity, following the FAO SDG 2.1.2 monitoring protocol ([FAO, 2014](#)):

$$\text{FIES}_{it} = \sum_{j=1}^8 y_{ijt}, \quad y_{ijt} \in \{0, 1\} \quad (4)$$

The binary indicators derived from HFIAS and FIES, together with a binary indicator of reduction in shock-induced food consumption, complete the set of results. The food reduction indicator is most directly related to the insurance hypothesis, as it captures the behavioral response to adverse shocks that remittances are theorized to mitigate.

2.3. Key Explanatory Variables

Two alternative remittance measures are used to identify different dimensions of remittance exposure. R1—HH Received Overseas Remittance (*hh_received_remitt0_1*) is a binary indicator equal to 1 if the household received any overseas remittance. This extensive-margin measure identifies whether remittances were received, and its IV estimate yields the Local Average Treatment Effect (LATE) for complier households. R2—Log Overseas Remittance Amount (*log_overseas_remittance*) is the natural log of the total overseas remittances (NRs.) received, defined only for households with positive receipts ($N = 6,470$; mean log amount = 9.89). This intensive-margin measure captures the income effect of remittance flows, conditional on households having received remittances.

The primary shock variable $S_{it} = \text{experienced_shock_s_it_1yes}$ is a binary indicator equal to 1 if the household experienced any adverse shock in the preceding twelve months (prevalence = 44.4%). The shock types are further disaggregated into natural shocks (28.4%), economic shocks (6.9%), health and death shocks (17.1%), covariate shocks (28.7%), and idiosyncratic shocks (21.8%) for heterogeneity analysis. All models include a vector of predetermined household controls X'_{it} : age and gender of the household head, household size, years of education, debt status, savings status, and public transfer receipt.

2.4. Identification Strategy: Shift-Share Bartik Instrumental Variable

The causal welfare effects of remittances are difficult to estimate due to a fundamental endogeneity problem. Households that send members abroad are positively selected in wealth, education, and social networks, all of which independently improve welfare (positive selection bias). Similarly, welfare deterioration can trigger increased migration, generating reverse causality. Both forces bias the OLS estimates in unpredictable directions. We address this using a Bartik (1991) shift-share instrument that generates exogenous variation in remittance exposure by interacting pre-determined district-level migration network composition with time-varying destination-country economic conditions. The instrument is constructed at the district-wave level as the exposure-weighted average of destination-country real GDP growth rates:

$$Z_{dt} = \sum_{k \in \mathcal{K}} s_{dk,2016} \times g_{k,t-1} \quad (5)$$

where Z_{dt} is the Bartik instrument for district d in year t ; $s_{dk,2016}$ is the baseline share of households in district d with a primary migrant destination in the country k as of the

2016 survey wave; and $g_{k,t-1}$ is the annual real GDP growth rate of the destination country k in the year prior to the survey wave (World Bank, 2023). The instrument spans four main destinations: Gulf states of the GCC (mean district share 24.6%), India (19.2%), Malaysia (13.2%) and other high-wage destinations (3.1%), with the remaining district share (39.9%) comprising households with no primary overseas migrant recorded as of the 2016 survey wave. The baseline shares are fixed at the 2016 values to avoid endogeneity of the shares themselves, consistent with the approach of Goldsmith-Pinkham et al. (2020). GDP growth rates are lagged one year to the survey reference period.

2.4.1. Justification of the Exclusion Restriction in the Nepal Context

Four features of Nepal’s external-sector structure support the assumption that destination-country GDP growth shocks affect Nepalese household welfare solely through remittance flows. First, trade integration with the identifying destinations is negligible: Nepal’s merchandise trade is overwhelmingly concentrated with India, with the GCC states and Malaysia collectively accounting for only a small share of exports and imports (World Bank, 2025). Second, foreign direct investment from migration-destination countries is marginal: India, China, Ireland, Singapore, and South Korea dominate Nepal’s FDI stock, while GCC states collectively hold less than 2% of approved commitments during the sample window (Department of Industry, 2025). Third, official development assistance from migration-destination countries is similarly limited: the main bilateral donors are India, China, Japan, the United Kingdom, and the United States, none of which overlap with the identifying destinations (Nepal Economic Forum, 2025). Fourth, tourism flows from the GCC and Malaysian nationals represent a negligible share of arrivals, preventing a tourism-driven transmission channel. Household and survey-year fixed effects further absorb time-invariant structural heterogeneity and nationwide shocks, isolating overseas remittances as the only economically meaningful channel of transmission.

2.5. Empirical Models

2.5.1. OLS Baseline

As a benchmark, we estimate the ordinary least squares model with household controls, household fixed effects, and year fixed effects:

$$W_{it} = \alpha + \gamma R_{it} + \beta S_{it} + \mathbf{X}_{it}'\boldsymbol{\delta} + \mu_i + \tau_t + \varepsilon_{it} \quad (6)$$

where W_{it} is the welfare outcome for the household i in wave t ; R_{it} is the remittance variable (R1 or R2); S_{it} is the shock indicator; μ_i represents households with fixed effects that absorb time-invariant household-level characteristics; τ_t denotes the fixed effects for the wave year; and ε_{it} is the idiosyncratic error term. The coefficient γ measures the conditional association between remittance receipt and welfare and is expected to be biased upward due to positive selection in migration. To maintain consistency with the district-wave variation of the Bartik instrument (Z_{dt}) and to account for spatial correlation between households within the same administrative unit, standard errors are clustered at the district level across all specifications.

2.5.2. Two-Stage Least Squares (2SLS) Direct Effect Model

To isolate the direct causal effect of remittances, the endogenous R_{it} is instrumented with Z_{dt} via two-stage least squares. The first stage is:

$$R_{it} = \pi_0 + \pi_1 Z_{dt} + \pi_2 S_{it} + \mathbf{X}'_{it} \boldsymbol{\theta} + \mu_i + \tau_t + \nu_{it} \quad (7)$$

The coefficient π_1 on Z_{dt} measures the relevance of the instrument. The strength of the first-stage is assessed using the Kleibergen–Paap rk Wald F -statistic (Kleibergen and Paap, 2006), which is robust to non-i.i.d. errors and is the appropriate diagnostic for 2SLS with heteroskedasticity of the disturbances.

The second stage substitutes the fitted values of the first-stage $\hat{R}_{it}$ into the structural equation:

$$W_{it} = \alpha + \gamma^{\text{IV}} \hat{R}_{it} + \beta S_{it} + \mathbf{X}'_{it} \boldsymbol{\delta} + \mu_i + \tau_t + \varepsilon_{it} \quad (8)$$

Under instrument relevance, exogeneity (Z_{dt} is not correlated with ε_{it}), and the exclusion restriction (Z_{dt} affects W_{it} only through R_{it}), the 2SLS estimator γ^{IV} is a consistent estimate of the Local Average Treatment Effect (LATE) for complier households (Angrist and Pischke, 2009).

2.5.3. IV-TWFE Buffering Interaction Model

The central contribution of the paper is to test whether overseas remittances buffer welfare losses induced by adverse shocks—the insurance hypothesis formalized in the New Economics of Labor Migration (Stark and Bloom, 1985; Lucas and Stark, 1985). The empirical model is a two-way fixed effects (TWFE) panel regression augmented with a remittance–shock interaction term:

$$W_{it} = \alpha + \gamma R_{it} + \beta S_{it} + \theta (R_{it} \times S_{it}) + \mathbf{X}'_{it} \boldsymbol{\delta} + \mu_i + \tau_t + \varepsilon_{it} \quad (9)$$

The key parameter of interest is θ , the coefficient of the remittance–shock interaction term. Under the insurance hypothesis, $\theta > 0$ for welfare-improving outcomes (FCS, HDDS, log food value, etc.), indicating that remittances attenuate the negative welfare impact of shocks, and $\theta < 0$ for welfare-deteriorating outcomes (HFIAS, FIES, food reduction, etc.), indicating that remittances reduce shock-induced food insecurity. The coefficient β measures the direct impact of shocks that hold remittances constant, and γ measures the direct effect of remittances that hold shocks constant.

The interaction term $R_{it} \times S_{it}$ is endogenous for the same reasons as R_{it} itself. We address this with a two-instrument IV strategy: Z_{dt} instruments for R_{it} , and $Z_{dt} \times S_{it}$ —the interaction of the Bartik instrument with the exogenous shock indicator—instruments for the endogenous interaction $R_{it} \times S_{it}$. This approach follows the standard treatment of endogenous interactions in IV models (Heckman and Vytlacil, 2005; Wooldridge, 2010) and is based on the logic that if Z_{dt} generates exogenous variation in R_{it} and S_{it} is exogenous, then $Z_{dt} \times S_{it}$ generates exogenous variation in the product $R_{it} \times S_{it}$. The two-equation first-stage system for the buffering model is:

$$R_{it} = \pi_0 + \pi_1 Z_{dt} + \pi_2 (Z_{dt} \times S_{it}) + \pi_3 S_{it} + \mathbf{X}'_{it} \boldsymbol{\theta}_1 + \mu_i + \tau_t + \nu_{1it} \quad (10)$$

$$(R_{it} \times S_{it}) = \lambda_0 + \lambda_1 Z_{dt} + \lambda_2 (Z_{dt} \times S_{it}) + \lambda_3 S_{it} + \mathbf{X}'_{it} \boldsymbol{\theta}_2 + \mu_i + \tau_t + \nu_{2it} \quad (11)$$

The fitted values $\hat{R}_{it}$ and $(\widehat{R_{it} \times S_{it}})$ of the first-stage equations are substituted into the second-stage equation to obtain consistent estimates of γ , β , and the buffering parameter θ .

2.5.4. Endogeneity Testing

To formally validate the use of IV estimation, we implement the Hausman endogeneity test (Hausman, 1978). Under the null hypothesis that R_{it} is exogenous, the OLS and IV estimators are consistent and converge. The test statistic is:

$$H = \frac{(\hat{\gamma}_{IV} - \hat{\gamma}_{OLS})^2}{\left| \widehat{\text{Var}}(\hat{\gamma}_{IV}) - \widehat{\text{Var}}(\hat{\gamma}_{OLS}) \right|} \quad (12)$$

which is asymptotically distributed as $\chi^2(1)$ under H_0 , with a 5% critical value of 3.84. Rejection of H_0 confirms endogeneity and the need for IV estimation. In the results, the Hausman statistic exceeds 3.84 for all welfare outcomes across both remittance measures in the smallholder sample, with values ranging from 4.33 to 68.81, providing strong evidence against exogeneity and validating the IV approach.

2.5.5. Heterogeneity and Subgroup Analysis

This study examines the heterogeneous effects of treatment in three dimensions. First, the buffering model (9) is estimated separately for the smallholder subsamples (≤ 0.5 ha) and non-smallholder subsamples (> 0.5 ha). Smallholder households are the main policy target given their narrow income base, greater dependence on migration, and limited access to formal insurance.

Second, the buffering model is re-estimated for each of ten shock types—natural shock, economic shock, health/death shock, covariate shock, idiosyncratic shock, drought, flood, price hike, illness, and death in the family—by replacing S_{it} with the type-specific indicator S_{it}^k and constructing new interaction instruments $Z_{dt} \times S_{it}^k$ accordingly. The shock-type specification is:

$$W_{it} = \gamma R_{it} + \sum_k \beta_k S_{it}^k + \sum_k \theta_k (R_{it} \times S_{it}^k) + \mathbf{X}'_{it} \boldsymbol{\delta} + \mu_i + \tau_t + \epsilon_{it} \quad (13)$$

Third, we estimate the full set of shock-by-outcome buffering effects: for each combination of shock type, welfare outcome, and remittance variable, we estimate θ , resulting in a comprehensive cross-analysis of insurance effectiveness across shock types and welfare dimensions.

2.5.6. Robustness Checks

To evaluate the consistency of the estimates and the strength of the identification strategy, multiple robustness checks are performed. First, reduced-form estimation is implemented by regressing each welfare outcome directly on Z_{dt} without the second stage. The

reduced-form coefficient should share the sign of the IV estimate and be statistically significant whenever the IV estimate is significant, since the Wald estimator identity implies $\gamma^{\text{IV}} = \hat{\pi}_{\text{RF}}/\hat{\pi}_1$. Second, OLS and IV estimates are presented side-by-side for all outcomes and remittance variables to document the direction and magnitude of endogeneity bias. Third, OLS and IV estimates of the buffering coefficient θ are compared; a genuine insurance effect should be strengthened rather than reversed by instrumentation. Fourth, as a non-parametric alternative identification strategy that does not rely on the Bartik instrument, we implement a kernel-weighted propensity score matched difference-in-differences (PSM-DiD) estimator; this is described in detail below.

2.5.7. Kernel-Weighted PSM-DiD Estimator

To validate the IV-2SLS estimates without relying on the shift-share Bartik instrument, we implement a kernel-weighted propensity score matched difference-in-differences (PSM-DiD) estimator. This approach combines propensity score matching—which controls for observable pre-treatment differences between remittance-receiving and non-receiving households—with a difference-in-differences structure that removes time-invariant unobserved heterogeneity, yielding a consistent estimate of the Average Treatment Effect on the Treated (ATT) under the conditional parallel trends assumption (Heckman et al., 1998; Blundell and Dias, 2009; Sant’Anna and Zhao, 2020; Callaway and Sant’Anna, 2021).

The estimation sample is limited to smallholder households only (≤ 0.5 ha), with both treated (smallholder households receiving overseas remittances) and control (smallholder households not receiving overseas remittances) drawn exclusively from this stratum. This design mirrors the main IV-2SLS smallholder subsample and ensures internal validity of the counterfactual: matched control units face the same land constraints, agro-ecological vulnerability, and shock exposure as treated units. The treatment variable is `hh_received_remitt0_1`, identical to `R1` in the IV-2SLS model. To verify that this restriction reflects genuine structural heterogeneity rather than sample selection, we re-estimate the identical PSM-DiD design on the full pooled sample and on non-smallholders only; these results are reported in the Appendix Tables A7 and A8 and discussed in Section 4.5.

Two complementary specifications are implemented, differing in the pre-post window. Specification A (2016 baseline) uses pre = Wave 2016 and post = mean(Waves 2017, 2018), restricted to $N = 4,287$ smallholder households (treated = 1,221; control = 3,066). Specification B (2017 baseline) uses pre = Wave 2017 and post = Wave 2018, with $N = 4,358$ (treated = 1,409; control = 2,949).

The DiD result for each specification is as follows:

$$\Delta W_i^A = \bar{W}_{i,\text{post}} - W_{i,2016} \quad (\text{Spec. A}), \quad \Delta W_i^B = W_{i,2018} - W_{i,2017} \quad (\text{Spec. B}) \quad (14)$$

The propensity score $P(T_i = 1 \mid X_i)$ is estimated within the smallholder subsample using binary logistic regression with L2 regularization, conditioning on all household controls from the IV-2SLS model (excluding the invariant smallholder indicator). The models achieve AUC = 0.717 (Spec. A) and AUC = 0.749 (Spec. B). The common support restriction retains 4,199 observations in Spec. A (support = [0.043, 0.751]) and 4,263 in Spec. B (support = [0.086, 0.824]).

The weights of the kernel are assigned using the Epanechnikov kernel (Heckman et al., 1998):

$$K(u) = 0.75(1 - u^2) \mathbf{1}\{|u| \leq 1\} \quad (15)$$

where $u = (p_i - p_j)/h$ is the bandwidth-scaled propensity score distance.

The kernel-weighted counterfactual for the treated unit i and the ATT estimator are given by:

$$\hat{y}_i^c = \frac{\sum_j K\left(\frac{p_i - p_j}{h}\right) \Delta W_j}{\sum_j K\left(\frac{p_i - p_j}{h}\right)}, \quad \widehat{\text{ATT}} = \frac{1}{N_1} \sum_{i: T_i=1} (\Delta W_i - \hat{y}_i^c) \quad (16)$$

where N_1 denotes the number of treated units within the region of common support.

The bandwidth h is selected using the Silverman (2018) rule-of-thumb,

$$h = 1.06 \hat{\sigma}(PS) N^{-1/5},$$

yielding $h = 0.0311$ (Spec. A) and $h = 0.0376$ (Spec. B).

Standard errors are computed using the analytical variance approximation (Abadie and Imbens, 2006):

$$\widehat{\text{Var}}(\widehat{\text{ATT}}) \approx \frac{\text{Var}(\Delta W_i - \hat{y}_i^c)}{N_1} \left(1 + \frac{N_1}{N_{c,\text{eff}}}\right) \quad (17)$$

where $N_{c,\text{eff}} = \min(N_0, 2N_1)$ denotes the effective number of control observations.

The bandwidth sensitivity is assessed over the range $h \times \{0.50, 0.75, 1.25, 1.50, 2.00\}$, with the results remaining stable across all specifications.

The PSM-DiD estimator is consistent under the conditional parallel trends assumption:

$$\mathbb{E}[\Delta W_i(0) \mid P(X_i), T_i = 1] = \mathbb{E}[\Delta W_i(0) \mid P(X_i), T_i = 0] \quad (18)$$

This assumption is supported by three considerations. First, restricting the sample to the smallholder stratum reduces structural heterogeneity in land endowments, income diversification, and exposure to shocks. Second, the rich set of covariates used to estimate the propensity score mitigates observable selection bias. Third, the difference-in-differences structure removes time-invariant unobserved confounders.

Finally, the consistency of ATT estimates in both sign and magnitude across baseline specifications provides an additional internal validity check. ATT is estimated for all welfare outcomes used in the IV-2SLS analysis, allowing a direct comparison between PSM-DiD and IV estimates across welfare dimensions.

3. Results

3.1. Descriptive Analysis: Welfare outcomes, Remittances, Shock exposure, and Household characteristics

Table 1 shows summary statistics of the Nepal Household Risk and Vulnerability Survey (HRVS, 2016–2018), covering 18,537 household-wave observations – 13,169 smallholders (≤ 0.5 ha, 71% of the sample) and 5,368 non-smallholders (> 0.5 ha). Group differences are

tested using one-way ANOVA F -tests for continuous variables and Pearson χ^2 tests for binary indicators.

This study finds that smallholder households face systematically worse welfare outcomes across all ten dimensions, with every difference significant at the 1-percent level. The logarithmic value of per capita food is lower for smallholders (5.932 vs. 6.061; $F = 291.31$; $p < 0.001$), as is the logarithmic annual food expenditure (11.400 vs. 11.614; $F = 741.65$; $p < 0.001$). Similarly, dietary quality, measured by the Food Consumption Score, is also significantly lower (81.152 vs. 86.122; $F = 181.85$; $p < 0.001$), while food insecurity access scores are higher (1.073 vs. 0.553; $F = 146.70$; $p < 0.001$). Subjective insecurity is similarly elevated among smallholders—13.1% face moderate-or-severe food insecurity compared to 7.3% of non-smallholders ($\chi^2 = 126.86$; $p < 0.001$), and smallholders are more likely to reduce food intake as a coping response to shocks (4.0 vs. 2.9%; $\chi^2 = 11.95$; $p < 0.001$).

Importantly, remittance receipt rates are statistically non-significant across the two groups (57.4 vs. 58.2%; $\chi^2 = 0.60$; $p = 0.439$), as are conditional overseas remittance amounts (9.880 vs. 9.916; $F = 0.14$; $p = 0.711$). This structural similarity in remittance exposure is critical for causal identification, as it rules out differential dosage across land-holding strata. The prevalence of aggregate shock is also similar (44.6 vs. 43.8%; $\chi^2 = 1.00$; $p = 0.317$), although smallholders face marginally higher economic shock rates (7.3 vs. 5.8%; $\chi^2 = 11.72$; $p < 0.001$).

Regarding household characteristics, smallholder-headed households are, on average, younger (48.07 vs. 50.80 years) and less educated (3.69 vs. 4.99 years; $p < 0.001$) than their non-smallholder counterparts. They are also significantly more likely to be female-headed (35.49 vs. 16.2%; $\chi^2 = 374.37$; $p < 0.001$) and less likely to belong to historically advantaged Brahmin/Chhetri castes (29.8 vs. 37.7%; $\chi^2 = 105.84$; $p < 0.001$). Smallholder households also face higher credit constraints (5.5 vs. 3.6%; $\chi^2 = 36.83$; $p < 0.001$) and cultivate significantly less land by construction (0.30 vs. 1.06 ha; $p < 0.001$).

The Bartik shift-share instrument averages 1.389 for smallholders versus 1.277 for non-smallholders ($F = 107.62$; $p < 0.001$), with a full-sample standard deviation of 0.667 providing sufficient district-wave variation to power first-stage regressions. Kleibergen–Paap F -statistics of 63.0 and 47.4 in the smallholder sample confirm the instrument’s reliability for identifying remittance impacts.

Appendix A1 breaks this comparison down by survey wave for the smallholder subsample, and the pattern is striking. Food insecurity falls sharply across the panel. The HFIAS score drops from 1.759 in 2016 to 1.120 in 2017, then just 0.328 by 2018 — an 81% decrease. FIES follows the same path, falling from 1.318 to 0.787 to 0.251. Dietary quality moves the other way, as expected: the Food Consumption Score rises from 78.9 to 84.0, and HDDS increases from 8.1 to 8.6. Log remittance amounts tell a complementary story, jumping from 8.16 in 2016 to roughly 10.8 in 2017–2018. This progression is consistent with the income-accumulation dynamic predicted by the NELM and aligns with the buffering results in Table 4 and the evidence of recovery-phase from the PSM-DiD estimates of Specification B in Table 5. The Bartik instrument itself declines mechanically in the panel, moving from 1.638 in 2016 to 1.409 in 2017 and 1.115 in 2018. This trajectory follows directly from how the instrument is built rather than signaling any real shift in migration patterns.

Table 1: Summary Statistics and Smallholder–Non-Smallholder Comparisons, Nepal HRVS (2016–2018)

Variable & Type	Full Sample		Smallholder (≤0.5 ha)		Non-Smallholder (>0.5 ha)		Diff. (SM–NS)	Statistic	<i>p</i> -value	Sig.
	N	Mean/%	N	Mean/%	N	Mean/%				
Panel A: Welfare Outcomes										
Log PC Food Value (7-day)	17,513	5.971	12,189	5.932	5,324	6.061	−0.129	F = 291.31	< 0.001	***
Log Annual Food Expenditure	18,054	11.464	12,686	11.400	5,368	11.614	−0.214	F = 741.65	< 0.001	***
Food Consumption Score (FCS)	18,056	82.629	12,688	81.152	5,368	86.122	−4.970	F = 181.85	< 0.001	***
Per Capita FCS	17,514	21.387	12,190	21.566	5,324	20.976	+0.590	F = 8.60	0.03	**
HFIAS Score (0–27)	18,056	0.919	12,688	1.073	5,368	0.553	+0.520	F = 146.70	< 0.001	***
FIES Score (0–8)	18,056	0.675	12,688	0.789	5,368	0.408	+0.380	F = 171.04	< 0.001	***
HDDS (0–12)	18,056	8.391	12,688	8.295	5,368	8.619	−0.324	F = 177.61	< 0.001	***
Any Food Insecure – HFIAS (0/1)	18,056	0.145	12,688	0.168	5,368	0.090	+7.8 pp	χ ² = 184.93	< 0.001	***
Mod/Sev Food Insecure – FIES (0/1)	18,056	0.114	12,688	0.131	5,368	0.073	+5.8 pp	χ ² = 126.86	< 0.001	***
Reduced Food Consumption (0/1)	18,221	0.037	12,921	0.040	5,300	0.029	+1.1 pp	χ ² = 11.95	< 0.001	***
Panel B: Remittances										
Received Overseas Remittance (0/1)	10,174	0.576	7,255	0.574	2,919	0.582	−0.8 pp	χ ² = 0.60	0.439	ns
Log Overseas Remittance Amount (NRs)	6,470	9.891	4,534	9.880	1,936	9.916	−0.036	F = 0.14	0.711	ns
Panel C: Shocks										
Experienced Any Shock (0/1)	18,221	0.444	12,921	0.446	5,300	0.438	+0.8 pp	χ ² = 1.00	0.317	ns
Natural Shock (0/1)	18,221	0.284	12,921	0.280	5,300	0.296	−1.6 pp	χ ² = 4.86	0.027	*
Economic Shock (0/1)	18,221	0.068	12,921	0.073	5,300	0.058	+1.4 pp	χ ² = 11.72	< 0.001	***
Panel D: Household Controls										
Age of HH Head (years)	17,701	49.503	12,333	48.070	5,368	50.797	−2.727	F = 244.15	0.031	**
Household Size	17,508	4.666	12,184	4.524	5,324	4.990	−0.466	F = 109.03	0.058	*
Years of Education (HH Head)	17,539	4.085	12,209	3.692	5,330	4.986	−1.295	F = 320.63	< 0.001	***
Panel E: Instrument										
Bartik IV (<i>Z_{dt}</i>)	18,056	1.356	12,688	1.389	5,368	1.277	+0.112	F = 107.62	< 0.001	***

Notes: Three survey waves (2016–2018) from the Nepal Household Risk and Vulnerability Survey (HRVS). SM = Smallholder (cultivated area ≤ 0.5 ha); NS = Non-Smallholder (cultivated area > 0.5 ha). For continuous variables, group means are compared using a one-way ANOVA F -test ($df = 1, N_{SM} + N_{NS} - 2$). For binary (0/1) variables, group proportions are compared using Pearson χ^2 tests of independence ($df = 1$, no continuity correction). Diff. (SM–NS): positive values indicate the smallholder mean/proportion is higher; negative values indicate the non-smallholder mean/proportion is higher. Differences for binary variables are expressed in percentage points (pp). The Bartik instrument (Z_{dt}) is a shift-share index of district-level remittance exposure constructed from national remittance growth rates. HFIAS = Household Food Insecurity Access Scale; FIES = Food Insecurity Experience Scale; HDDS = Household Dietary Diversity Score; FCS = Food Consumption Score; HH = Household. All statistics are computed from the verified final dataset ($N = 18,537$ household-wave observations).

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$, ns = not significant.

3.2. Endogeneity Validation and Instrument Strength

Before presenting causal estimates, the validity of the instrumental variable approach is confirmed by two sets of diagnostics. First, Hausman endogeneity tests (Appendix A2) reject the nullity of OLS consistency for both remittance measures in the smallholder sample, with $\chi^2(1)$ statistics ranging from 4.33 (per capita FCS, $R1$) to 68.81 (HDDS, $R1$), all exceeding the 5% critical value of 3.84. This decisively establishes that both the receipt of remittance ($R1$) and the amount of log remittance ($R2$) are endogenous in the welfare equations, consistent with positive selection into migration and reverse causality concerns documented in the broader migration–welfare literature (Mora-Rivera and van Gasteren, 2021; Jayaweera and Verma, 2024).

Second, first-stage Kleibergen–Paap F -statistics confirm strong instrument relevance for the direct models: $F = 63.0$ ($R1$) and $F = 47.4$ ($R2$), both substantially exceeding the Stock–Yogo critical value of 16.38 (Appendix A3). For the two-instrument buffering system, which employs both Z_{dt} and $Z_{dt} \times S_{it}$ as instruments, F -statistics are 27.3 ($R1$) and 16.7 ($R2$). The $R2$ buffering model marginally clears the weak-instrument threshold and is therefore interpreted with caution, following standard guidance on weak-instrument diagnostics (Borusyak et al., 2022). The reduced-form estimates (Appendix A4) confirm that the Bartik instrument directly shifts all welfare outcomes in the expected direction, validating the Wald estimator identity and the instrument’s theoretical mechanism.

3.3. OLS Estimates and Direction of Endogeneity Bias

Table 2 reports the baseline estimates for fixed effects of OLS for both remittance measures across the smallholder subsample and throughout the sample. Across all ten welfare outcomes, the OLS coefficients on overseas remittance receipt ($R1$) and log remittance amount ($R2$) are consistently signed in the expected direction, providing a preliminary picture of the association between remittances and household welfare.

For food security outcomes, remittance receipt is negatively associated with food insecurity: the OLS coefficient on $R2$ is -0.082 for the HFIAS score ($p < 0.001$) and -0.051 for the FIES score ($p < 0.001$) in the smallholder sample, while the binary HFIAS food insecurity indicator falls by 1.1 percentage points and the moderate-or-severe FIES indicator by 0.9 percentage points per log-unit increase in remittances.

For the dietary quality outcomes, remittances are positively associated with welfare: the OLS coefficient on $R2$ is 0.807 for the Food Consumption Score ($p < 0.001$), 0.064 for the HDDS ($p < 0.001$), and 0.036 for log per capita food value ($p = 0.016$). The OLS estimates for the full sample align closely with the smallholder-specific results, confirming that the directional patterns are not an artifact of subsample composition.

Shocks, captured by the coefficient β in the shock indicator S_{it} , are strongly and significantly associated with worse outcomes across all dimensions: $\beta = 0.506$ for HFIAS ($p < 0.001$), 0.360 for FIES ($p < 0.001$), and -0.293 for HDDS ($p = 0.002$), confirming the severity of deterioration of welfare induced by shocks even after conditioning on household and year fixed effects. Despite these consistent directional patterns, OLS estimates cannot be given a causal interpretation because remittance receipt is endogenous to household welfare. Positive selection into migration—where wealthier and better-connected households are

more likely to send members abroad—generates upward bias in the OLS coefficient on the remittance variable. At the same time, the labor-withdrawal channel of the New Economics of Labor Migration operates in the opposite direction: households that send productive members abroad simultaneously lose on-farm labor, which temporarily depresses food production and consumption before remittance flows fully compensate. The net direction of the OLS bias is therefore ambiguous *a priori* and outcome-specific.

Comparing the OLS and IV estimates reveals this bias clearly: for the HFIAS score, the OLS coefficient on $R2$ is -0.082 , while the IV estimate is -0.825 —a tenfold difference that indicates severe attenuation of the true causal effect by the labor-withdrawal mechanism. For dietary quality outcomes such as the Food Consumption Score, the OLS coefficient of 0.807 underestimates the IV estimate of 6.41 , again consistent with the labor-withdrawal channel compressing the OLS estimate toward zero.

Hausman endogeneity tests formally confirm these concerns: the test statistic $\chi^2(1)$ exceeds the critical value of 5% of 3.84 for all ten welfare outcomes under both remittance measures in the smallholder sample, with values ranging from 4.33 (per capita FCS, $R1$) to 68.81 (HDDS, $R1$). These results establish that the OLS estimates are inconsistent and validate the use of the Bartik instrumental variable strategy for all subsequent causal inferences.

Table 2: OLS Baseline Estimates: All Samples, R1 and R2

Welfare Outcome	R1 — Smallholder (Binary Receipt)						R2 — Smallholder (Log Amount)						R2 — Full Sample (Log Amount)					
	γ	SE	β	SE	p	Sig.	γ	SE	β	SE	p	Sig.	γ	SE	β	SE	p	Sig.
Log PC Food Value (7-day)	0.05	(0.02)	-0.08	(0.02)	<0.01	***	0.04	(0.00)	-0.07	(0.00)	0.02	**	0.00	(0.00)	-0.06	(0.00)	0.04	**
Log Annual Food Expenditure	0.03	(0.02)	-0.09	(0.02)	0.07	*	0.01	(0.00)	-0.09	(0.00)	<0.01	***	0.01	(0.00)	-0.07	(0.00)	<0.01	***
Food Consumption Score (FCS)	2.62	(0.91)	-2.17	(0.91)	0.00	***	0.81	(0.12)	-1.26	(0.12)	<0.01	***	0.57	(0.09)	-0.58	(0.09)	<0.01	***
Per Capita FCS	1.13	(0.38)	-0.48	(0.38)	0.00	***	0.07	(0.06)	-0.13	(0.06)	0.24	ns	0.02	(0.04)	-0.02	(0.04)	0.60	ns
HFIAS Score	-0.48	(0.12)	0.51	(0.12)	<0.01	***	-0.08	(0.02)	0.57	(0.02)	<0.01	***	-0.06	(0.01)	0.56	(0.01)	<0.01	***
FIES Score (0-8)	-0.30	(0.08)	0.36	(0.08)	<0.01	***	-0.05	(0.01)	0.35	(0.01)	<0.01	***	-0.04	(0.01)	0.35	(0.01)	<0.01	***
HDDS Score (0-12)	0.18	(0.06)	-0.29	(0.06)	0.00	***	0.06	(0.01)	-0.14	(0.01)	<0.01	***	0.05	(0.01)	-0.08	(0.01)	<0.01	***
Any Food Insecure – HFIAS (0/1)	-0.05	(0.02)	0.07	(0.02)	0.02	**	-0.01	(0.00)	0.07	(0.00)	<0.01	***	-0.01	(0.00)	0.07	(0.00)	<0.01	***
Mod/Sev Food Insecure – FIES (0/1)	-0.05	(0.01)	0.06	(0.01)	<0.01	***	-0.01	(0.00)	0.05	(0.00)	<0.01	***	-0.01	(0.00)	0.05	(0.00)	<0.01	***
Reduced Food Consumption (0/1)	0.03	(0.01)	0.09	(0.01)	0.00	***	0.00	(0.00)	0.09	(0.00)	0.24	ns	0.00	(0.00)	0.09	(0.00)	0.36	ns
N (median)	$\approx 6,073$						$\approx 4,209$						$\approx 6,080$					
Estimator	OLS-FE						OLS-FE						OLS-FE					
HH Fixed Effects	Yes						Yes						Yes					
Year Fixed Effects	Yes						Yes						Yes					
Within R^2	0.09–0.28						0.11–0.36						0.04–0.17					

Notes: OLS with household (HH) and year fixed effects. Model: $W_{it} = \alpha + \gamma R_{it} + \beta S_{it} + X'_{it}\delta + \mu_i + \tau_t + \varepsilon_{it}$. γ is upward biased due to positive selection into migration. $R1$ = binary overseas remittance receipt; $R2$ = log overseas remittance amount. β = coefficient on shock indicator S_{it} . All numeric values rounded to two decimal places. Controls include age and gender of head, household size, years of education, ethnicity, debt status, savings status, and public transfer receipt. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; ns = not significant.

3.4. IV-2SLS Direct Effects of Overseas Remittances on Household Welfare

Table 3 presents two-stage least squares (2SLS) estimates of how overseas remittances causally affect ten welfare outcomes under three specifications: binary remittance receipt (R1) for the smallholder subsample, log overseas remittance amount (R2) for the smallholder subsample, and R2 for the full sample. Before turning to the estimates themselves, it is worth noting that the instrument strength is robust across all specifications. The Kleibergen–Paap F-statistics are 63.0 (R1, smallholder), 47.4 (R2, smallholder), and 48.9 (R2, full sample)—all far exceeding the Stock–Yogo critical value of 16.38 for 10% maximal IV size distortion. Hausman endogeneity tests confirm that OLS is inconsistent for all outcomes, validating the IV approach throughout.

The most pronounced differences emerge in indicators of food insecurity. Among smallholder households, an increase of one-log unit in overseas remittances, equivalent to a substantial increase in the amount received, reduces the HFIAS score by 0.825 points and the FIES score by 0.610 points (both $p < 0.001$). This effect is substantial in magnitude: relative to the sample mean HFIAS score of 1.073, a one-log unit increase in remittances is associated with a reduction of approximately 50% in experienced food insecurity. Binary indicators convey the same result in more intuitive terms: the probability of being food insecure under the HFIAS falls by 15.9 percentage points, and the probability of falling into moderate-or-severe insecurity under the FIES falls by 10.6 percentage points (both $p < 0.001$). These effects are not limited to the margins of the distribution; they reflect a substantively significant improvement in the daily welfare of a large share of Nepal’s most vulnerable agricultural households. Perhaps the most telling result concerns the indicator of reduction in food consumption: the probability that a household cuts meals as a coping response to shocks decreases by 6.4 percentage points ($p < 0.001$).

Remittances also improve the broader dimensions of dietary quality. The Food Consumption Score shows a marginally significant increase of 6.408 points ($p = 0.063$), and the Household Dietary Diversity Score shows a marginally significant improvement of 0.650 points ($p = 0.092$). Log per capita food value increases by 0.098 points ($p = 0.047$) and log annual food expenditure by a marginally significant 0.077 points ($p = 0.072$). The pattern here is actually quite interpretable: when additional income arrives in a household that has been struggling with food access, the first priority is to stop being hungry, not to diversify the plate. Dietary diversity is a second-order concern, and weaker effects on FCS and HDDS simply reflect that order of priorities. It should also be noted that Nepal’s nationally high FCS baseline (mean of 82.6 in the full sample, comfortably in the ‘acceptable’ range) leaves limited statistical room for significant improvements in measures of dietary quality, even when remittance income is substantial.

The two remittance measures tell a consistent directional story but differ in scale in a way that is theoretically informative. Under the binary receipt indicator (R1), which captures the Local Average Treatment Effect for households that move from receiving nothing to receiving something, the food insecurity effects are dramatically greater: HFIAS decreases by 5.210 points ($p < 0.001$) and FIES by 3.772 points ($p < 0.001$). These large estimates of R1 reflect what is really a composite: the income gain from remittances, but also the shift in household circumstances that comes with having an overseas migrant in the first place.

The log-amount measure (R2), which is conditional on receipt and asks what an additional unit of remittance income does at the margin, gives more conservative and arguably more policy-useful estimates.

Table 3: IV-2SLS Direct Effects of Overseas Remittances on Household Welfare

Welfare Outcome	<i>R1: Binary Receipt — Smallholder</i>			<i>R2: Log Overseas Amount — Smallholder</i>			<i>R2: Log Overseas Amount — Full Sample</i>		
	γ	Std. Error	p-value	γ	Std. Error	p-value	γ	Std. Error	p-value
Log PC Food Value (7-day)	0.137*	(0.075)	0.068	0.098**	(0.049)	0.047	0.142**	(0.063)	0.025
Log Annual Food Expenditure	0.089**	(0.042)	0.036	0.077*	(0.043)	0.072	0.120**	(0.060)	0.047
Food Consumption Score (FCS)	17.359	(13.09)	0.185	6.408*	(3.448)	0.063	8.931	(7.731)	0.248
Per Capita FCS	3.746	(1.805)	0.339	1.155	(0.753)	0.125	1.272	(0.456)	0.377
HFIAS Score	-5.210***	(1.179)	< 0.001	-0.825***	(0.139)	< 0.001	-0.806***	(0.122)	< 0.001
FIES Score (0–8)	-3.772***	(0.764)	< 0.001	-0.610***	(0.090)	< 0.001	-0.588***	(0.080)	< 0.001
HDDS Score (0–12)	4.688	(3.153)	0.137	0.650*	(0.386)	0.092	0.667*	(0.350)	0.057
Any Food Insecure – HFIAS (0/1)	-0.257*	(0.153)	0.094	-0.159***	(0.018)	< 0.001	-0.159***	(0.017)	< 0.001
Mod/Sev Food Insecure – FIES (0/1)	-0.171*	(0.097)	0.078	-0.106***	(0.016)	< 0.001	-0.101***	(0.015)	< 0.001
Reduced Food Consumption (0/1)	-0.401***	(0.098)	< 0.001	-0.064***	(0.012)	< 0.001	-0.064***	(0.011)	< 0.001
KP F-statistic (1st stage)	63.0			47.4			48.9		
N (median across outcomes)	$\approx 6,073$			$\approx 4,209$			$\approx 6,080$		
HH Fixed Effects	Yes			Yes			Yes		
Year Fixed Effects	Yes			Yes			Yes		
HH Controls	Yes			Yes			Yes		

Notes: Two-stage least squares (2SLS). Model: $W_{it} = \alpha + \gamma R_{it} + \beta S_{it} + X'_{it}\delta + \mu_i + \tau_t + \varepsilon_{it}$, instrumented by Z_{bartik} . γ = causal effect of remittance on welfare. $R1$ = binary overseas remittance receipt; $R2$ = log overseas remittance amount. IV identification: The Bartik shift-share instrument (Z_{dt}) exploits plausibly exogenous variation in district-level remittance exposure driven by destination-country GDP growth rates interacted with pre-2016 migrant network shares. R^2 is not reported for IV/2SLS models as it can be misleading; instrument strength is assessed via the Kleibergen–Paap (KP) F-statistic. Stock–Yogo critical value: 16.38 (10% maximal IV size distortion). Fixed effects & controls: All models include HH fixed effects and survey-year fixed effects. Time-varying household controls: age and gender of head, household size, years of education, ethnicity, debt status, savings status, and public transfer receipt. All numeric values rounded to three decimal places. *Significance*: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.5. IV-TWFE Buffering Model: Remittances as Informal Insurance

The direct effects in Table 3 tell us that remittances improve welfare on average. But the more important question for Nepal’s shock-prone smallholder households is whether remittances provide protection when things go wrong. Table 4 addresses that question directly. To evaluate how remittances mitigate adverse conditions, the IV two-way fixed effects model introduces a remittance–shock interaction term. This yields three primary coefficients of interest: γ , representing the direct effect of remittances in the absence of a shock; β , capturing the welfare loss experienced by unremitted households during a shock; and θ , the buffering coefficient. The parameter θ serves as our primary focus, as it quantifies the extent to which remittances offset shock-induced welfare losses. Table 4 presents both remittance measures side by side: the binary receipt indicator (R1, KP-F = 27.3) and the log amount (R2, KP-F = 16.7). Both comfortably exceed the Stock–Yogo threshold of 16.38, although the R2 buffering system sits closer to the boundary and warrants slightly more cautious interpretation (Borusyak et al., 2022).

The shock coefficients (β) in Table 4 confirm the severity of adverse shocks in smallholder households. Under R1 (R2), a shock increases the HFIAS score by 2.35 (5.501) points, increases the FIES score by 6.239 (4.005) points, and increases the probability of food insecurity by 0.871 (0.918) percentage points. Under the continuous remittance measure (R2), the buffering coefficient (θ) on HFIAS is -1.045 ($p = 0.005$), indicating that a one-log-unit increase in overseas remittances offsets approximately 19% of the shock-induced deterioration in HFIAS, which otherwise averages 5.501 points. For the FIES score, $\theta = -0.753$ ($p = 0.001$), again offsetting approximately 19% of the shock-induced deterioration in FIES, which otherwise averages 4.005 points. Binary food insecurity indicators confirm this pattern with results that are easy to interpret in practical terms: households receiving remittances are 0.202 percentage points less likely to cross the HFIAS threshold for any-insecurity after a shock ($p < 0.001$), and 0.101 percentage points less likely to experience moderate-or-severe insecurity under FIES ($p = 0.016$). For the food consumption reduction indicator—the outcome that captures households cutting meals as a last resort— $\theta = -0.118$ ($p = 0.031$) under R2, which more than offsets the direct shock effect of 0.083. After an adverse shock, smallholder households receiving remittances exhibit a markedly reduced likelihood of reducing meal frequency as a coping strategy.

The R1 estimates in the left panel of Table 4 use the binary remittance receipt indicator and are identified with a stronger instrument (KP-F = 27.3). They tell the same story, but with considerably larger magnitudes, since R1 captures the full transition from no remittance income to positive remittance income. For the HFIAS score, $\theta = -8.830$ ($p < 0.001$), and for the FIES, $\theta = -1.660$ ($p < 0.001$). The any-HFIAS-insecurity buffering coefficient is -0.499 ($p = 0.083$), and the food consumption reduction indicator is buffered by -0.274 ($p = 0.056$). These larger R1 estimates reflect the LATE for households that cross the threshold from non-receipt to receipt, a group for whom gaining access to any remittance income represents a fundamentally different welfare position. One notable divergence under R1 is that log annual food expenditure shows a positive and significant buffering effect ($\theta = 1.975$; $p = 0.038$), and the FCS shows a marginally significant positive θ ($p = 0.084$), suggesting that at the extensive margin, households that begin receiving remittances during a shock period can

actually maintain or even improve their food spending and dietary quality—not merely avoid the worst outcomes.

Not every buffering effect is statistically significant, and this pattern of null results is itself informative. The buffering coefficients for log per capita food value, log annual food expenditure, FCS, per capita FCS, and HDDS are all statistically indistinguishable from zero under R2. This does not mean that remittances fail to protect households—it means that during a shock episode, the primary function of remittance income is to prevent acute food insecurity rather than to maintain or upgrade dietary diversity. During episodes of acute shock exposure, such as drought or flooding, households prioritize ensuring minimum food access over dietary diversification. Remittances help achieve that, as the HFIAS, FIES, and food reduction results clearly show. But the more aspirational goal of sustaining a diverse, nutritious diet, captured by the FCS and HDDS, is a second-order concern during a crisis, and remittances are not a sufficiently large income source to buffer that dimension simultaneously.

Table 4: IV-TWFE Buffering Model: Full Detail (γ , β , θ): R1 and R2, and Smallholder Households

Welfare Outcome	R1 — Smallholder (Binary Receipt) KP-F = 27.3, N $\approx$ 6,073								R2 — Smallholder (Log Amount) KP-F = 16.7, N $\approx$ 4,209							
	γ	SE(γ)	β	SE(β)	θ	SE(θ)	p(θ)	Sig.	γ	SE(γ)	β	SE(β)	θ	SE(θ)	p(θ)	Sig.
Log PC Food Value (7-day)	0.365	(0.163)	−1.909	(0.528)	0.044	(0.997)	0.684	ns	0.095	(0.019)	−1.336	(0.018)	0.048	(0.049)	0.336	ns
Log Annual Food Expenditure	0.148	(0.022)	−0.918	(0.503)	1.975	(0.951)	0.038	**	0.078	(0.018)	−3.832	(0.131)	0.018	(0.047)	0.496	ns
Food Consumption Score (FCS)	23.900	(7.140)	−13.50	(6.920)	18.539	(11.650)	0.084	*	6.325	(1.027)	−3.992	(2.479)	1.291	(0.597)	0.219	ns
Per Capita FCS	4.470	(1.230)	−7.518	(2.370)	−15.750	(10.034)	0.117	ns	1.142	(0.474)	−13.090	(3.381)	−0.202	(1.236)	0.870	ns
HFIAS Score	−8.450	(2.711)	2.350	(1.461)	−8.830	(3.375)	< 0.001	***	−0.757	(0.122)	5.501	(0.808)	−1.045	(0.369)	0.005	***
FIES Score (0–8)	−9.431	(3.691)	6.239	(2.227)	−1.660	(0.577)	< 0.001	***	−0.561	(0.080)	4.005	(0.547)	−0.753	(0.236)	0.001	***
HDDS Score (0–12)	7.480	(1.113)	−11.330	(1.908)	11.710	(3.604)	0.095	*	0.646	(0.068)	−1.930	(0.490)	0.062	(0.175)	0.725	ns
Any Food Insecure – HFIAS (0/1)	−0.480	(0.266)	0.871	(0.465)	−0.499	(0.859)	0.083	*	−0.146	(0.017)	0.918	(0.115)	−0.202	(0.047)	< 0.001	***
Mod/Sev Food Insecure – FIES (0/1)	−0.301	(0.085)	0.799	(0.414)	−0.189	(0.763)	0.247	ns	−0.099	(0.015)	0.772	(0.103)	−0.101	(0.042)	0.016	**
Reduced Food Consumption (0/1)	−0.749	(0.090)	0.512	(0.213)	−0.274	(0.140)	0.056	**	−0.767	(0.068)	0.083	(0.031)	−0.118	(0.038)	0.031	**
KP F-stat (buffering)	27.3								16.7							
N	$\approx$ 6,073								$\approx$ 4,209							
HH Fixed Effects	Yes								Yes							
Year Fixed Effects	Yes								Yes							
HH Controls	Yes								Yes							

Notes: Full IV-TWFE buffering model: $W_{it} = \alpha + \gamma R_{it} + \beta S_{it} + \theta(R_{it} \times S_{it}) + X'_{it}\delta + \mu_i + \tau_t + \varepsilon_{it}$. Two instruments: Z_{bartik} (for R_{it}) and $Z_{\text{bartik}} \times S_{it}$ (for $R_{it} \times S_{it}$). γ = direct remittance effect; β = direct shock effect; θ = key buffering parameter. Insurance hypothesis: $\theta < 0$ for food insecurity outcomes (HFIAS, FIES, food reduction); $\theta > 0$ for dietary quality outcomes (FCS, HDDS, food value). R1 = binary remittance receipt (KP-F = 27.3); R2 = log overseas remittance amount (KP-F = 16.7, use with caution). Both specifications: smallholder subsample (≤ 0.5 ha), HH and year fixed effects, all household controls. The Bartik shift-share instrument (Z_{dt}) exploits plausibly exogenous variation in district-level remittance exposure from destination-country GDP growth rates interacted with pre-2016 migrant network shares. Stock–Yogo critical value: 16.38 (10% maximal IV size distortion). All values rounded to three decimal places. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; ns = not significant.

3.6. Robustness Checks: PSM-DiD

The Bartik shift-share design rests on the assumption that fluctuations in destination-country GDP growth are uncorrelated with unobserved household welfare determinants. To assess the robustness of this assumption, Table 5 reports results from an entirely different identification strategy that does not rely on the Bartik instrument: a kernel-weighted propensity score matched difference-in-differences (PSM-DiD) estimator. This approach constructs a control group of smallholder households with observable characteristics similar to remittance-receiving households, then applies a difference-in-differences structure to remove remaining time-invariant unobserved heterogeneity. Both treated and control households are drawn from the smallholder stratum, differing only in overseas remittance receipt. Propensity scores are estimated by L2-regularized logistic regression with AUC values of 0.717 (Specification A) and 0.749 (Specification B), and common support is maintained across the full range of propensity scores, giving confidence that the matched households are genuinely comparable (Caliendo and Kopeinig, 2008; Heckman et al., 1998).

The two baseline specifications are not redundant; they trace different phases of the welfare trajectory predicted by the New Economics of Labor Migration (NELM). Specification A (2016 baseline) compares the results in the average of waves 2017–2018 against the Wave 2016 baseline, capturing the medium-term welfare change that spans the initial adjustment of the household’s to having a member abroad. Specification B (2017 baseline) isolates only the Wave 2018 outcome against the Wave 2017 baseline, capturing what happens in the terminal year once remittances have been flowing for a full migration cycle. Together, these two specifications provide a dynamic picture of how remittances affect welfare over time, a dimension that cross-sectional IV-2SLS estimates cannot capture (Callaway and Sant’Anna, 2021).

3.6.1. Specification A: The Medium-Run Picture

Specification A estimates reveal that smallholder households receiving remittances are significantly less likely to reduce food consumption as a coping response over the medium-term window, a result with direct implications for household welfare during crises ($ATT = -0.018$; $t = -1.726$; $p = 0.084$). This corroborates the IV-2SLS finding and confirms, using a non-parametric strategy independent of the instrument, that remittances protect households from this last-resort coping behavior. In addition, log annual food expenditure is significantly higher for remittance-receiving households over the medium run ($ATT = +0.042$; $p = 0.006$), providing direct evidence that the income effect of remittances is not merely theoretical but translates into actual food spending. Most other outcomes, including FCS, HDDS, and log per capita food value, show positive but statistically insignificant ATT estimates. This pattern aligns with the predictions of NELM for the transition period, where the labor-withdrawal cost of sending a productive member abroad partially offsets the income gain, dampening the net improvement in welfare in the medium term and keeping it below significance for the more diffuse dietary quality indicators (Paudel et al., 2025).

3.6.2. *Specification B: The Recovery Phase Emerges*

Specification B captures the post-adjustment period, once remittance flows have had sufficient time to accumulate. The per capita FCS shows a positive and marginally significant ATT of +0.376 ($p = 0.051$), suggesting that by the end of the year, households receiving remittances are beginning to improve the quality and diversity of their diet in ways that were not visible in the medium-run window. This trajectory, from weak initial effects on dietary quality to a positive signal by the panel's end, is consistent with the recovery dynamic predicted by NELM as remittances accumulate and household income strengthens. The food insecurity indicators under Specification B (HFIAS: ATT = -0.050 , $p = 0.606$; FIES: ATT = -0.037 , $p = 0.560$) move in the expected direction but do not reach statistical significance, likely because the most acute food insecurity has already been addressed in the earlier period and the remaining variation is smaller. Log annual food expenditure under Specification B is near zero and insignificant (ATT = $+0.002$; $p = 0.261$), consistent with a welfare trajectory that has already stabilized by Wave 2017. Together, the two specifications corroborate the IV-TWFE buffering results, reinforcing that overseas remittances function as an informal insurance mechanism for Nepalese smallholder households—first as a floor against welfare deterioration, then, as migration matures, as a platform for dietary recovery.

Table 5: Robustness Check: Kernel-Weighted PSM-DiD–Smallholder Households Only

Welfare Outcome	Specification A — 2016 Baseline						Specification B — 2017 Baseline					
	ATT	SE	<i>t</i> -stat	<i>p</i> -val	CI _{lo}	CI _{hi}	ATT	SE	<i>t</i> -stat	<i>p</i> -val	CI _{lo}	CI _{hi}
Log PC Food Value (7-day)	0.015	(0.016)	0.942	0.347	−0.016	0.045	0.026	(0.017)	1.582	0.115	−0.006	0.058
Log Annual Food Expenditure	0.042	(0.015)	2.747	0.006	0.012	0.073	0.002	(0.002)	1.125	0.261	−0.030	0.034
Food Consumption Score (FCS)	0.116	(0.878)	0.132	0.895	−1.605	1.836	0.540	(0.900)	0.599	0.549	−1.225	2.305
Per Capita FCS	—	—	—	—	—	—	0.376	(0.193)	1.949	0.051	0.088	0.663
HFIAS Score	—	—	—	—	—	—	−0.050	(0.097)	−0.516	0.606	−0.239	0.139
FIES Score (0–8)	—	—	—	—	—	—	−0.037	(0.063)	−0.583	0.560	−0.161	0.087
HDDS Score (0–12)	0.008	(0.053)	0.140	0.889	−0.097	0.112	0.021	(0.057)	0.376	0.706	−0.090	0.132
Any Food Insecure – HFIAS (0/1)	—	—	—	—	—	—	−0.014	(0.013)	−1.109	0.268	−0.039	0.011
Mod/Sev Insecure – FIES (0/1)	—	—	—	—	—	—	−0.008	(0.011)	−0.721	0.472	−0.030	0.014
Reduced Food Consumption (0/1)	−0.018	(0.011)	−1.726	0.084	−0.039	0.003	0.002	(0.008)	0.195	0.846	−0.014	0.017
<i>N</i> (treated / control)	1,221 / 3,066						1,409 / 2,949					
Kernel / Bandwidth	Epanechnikov / 0.031						Epanechnikov / 0.038					
AUC (PS model)	0.717						0.749					
Common Support	[0.043, 0.751]						[0.086, 0.824]					
HH Fixed Effects	Yes						Yes					
Year Fixed Effects	Yes						Yes					

Notes: ATT = Average Treatment Effect on the Treated (Epanechnikov kernel-weighted, Silverman bandwidth). Treatment: smallholder household received overseas remittance. Control: smallholder household did not receive remittances. Both groups restricted to the smallholder stratum (≤ 0.5 ha). DiD outcome: Spec. A = $\bar{W}_{\text{post}} - W_{2016}$ (medium-run); Spec. B = $W_{2018} - W_{2017}$ (recovery phase). Propensity scores estimated by L2-regularised logistic regression on all household controls: age and gender of head, household size, years of education, ethnicity, debt status, savings status, and public transfer receipt. Standard errors from Abadie–Imbens (2006); 95% CIs reported. Dashes (—) = outcome not estimated in Specification A. All values rounded to three decimal places. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4. Discussion

4.1. Remittances Reduce Baseline Food Insecurity: What the Direct Effects Tell Us

The IV-2SLS estimates in Table 3 resolve a long-standing ambiguity in the Nepalese migration literature: once the endogeneity of migration is considered, overseas remittances deliver large and causally identified improvements in household food security. An increase in a log-unit in overseas remittance amounts reduces the HFIAS score by 0.825 points ($p < 0.001$) and the FIES score by 0.610 points ($p < 0.001$). To place this in context, the smallholder mean HFIAS is 1.073, so a single log-unit increase in remittances essentially halves the average score. The probability of crossing the any-insecurity threshold falls by 15.9 percentage points, and the probability of moderate-or severe food insecurity falls by 10.6 percentage points. These estimates represent a substantive shift in food security outcomes for a considerable share of affected households, rather than marginal changes confined to the tail of the distribution. By contrast, indicators related to diet quality show suggestive but not conclusive responses: the Food Consumption Score increases by 6.41 points ($p = 0.063$) and the Household Dietary Diversity Score by 0.650 ($p = 0.092$), both of which fall short of the conventional threshold. This pattern is consistent with the broader literature, which finds that remittances more reliably improve food access and consumption smoothing than dietary diversity or nutritional quality specifically (Thow et al., 2016; Zingwe et al., 2023). The food consumption reduction indicator, the outcome that most directly captures coping behavior, falls by 6.4 percentage points, indicating that smallholder households receiving remittances are substantially less likely to reduce meal frequency during periods of hardship.

These results are consistent with the growing body of instrumental variable (IV) research conducted in similar low-income contexts recently. Mora-Rivera and van Gasteren (2021) employ ordered probit IV models with Mexican rural households and find that international remittances significantly reduce HFIAS scores, providing some of the first causal evidence on food insecurity specifically. Alam et al. (2025) draws on a national survey in Bangladesh and finds that remittance receipt reduces food insecurity by more than two-fold on an ATT basis, a result that mirrors the large magnitude we document for the smallholder subsample. Jayaweera and Verma (2024) find positive impacts on food expenditure and calorie consumption in Sri Lanka using propensity score matching, and Zingwe et al. (2023) document insurance-type effects in Malawi in the context of multiple concurrent shocks. Our estimates are consistent in direction with all of these studies and broadly comparable in magnitude to their IV-based results, a convergence that is reassuring given that our identification strategy is distinct from the instruments used elsewhere. The convergence of directional findings across such varied empirical approaches and contexts makes it difficult to dismiss the result as a local statistical artifact. What this paper contributes is the first quasi-experimental panel estimate for Nepal specifically, a context in which remittances now represent 26.6% of GDP and the welfare stakes of getting the causal question right are unusually high (International Organization for Migration, 2024; Thapa-Parajuli et al., 2025).

The observed prioritization of food quantity over dietary diversity reflects a fundamental economic behavior in developing regions around the world. In lower and middle-income

countries characterized by severe liquidity constraints and absent formal credit markets, remittance transfers act primarily as downside risk protection, alleviating immediate calorie deficits and reducing acute food insecurity before households allocate marginal migration income toward nutritional quality or food group diversification (Alam et al., 2025; Roshania and Haardörfer, 2025).

4.2. *The OLS Bias and What It Reveals About Mechanism*

One of the more instructive results in this paper is not the IV estimate itself but the gap between the IV and the OLS. For the HFIAS score, the OLS coefficient on the amount of logarithmic remittance is -0.082 , while the IV estimate is -0.825 . For the Food Consumption Score, the OLS gives 0.807 while the IV gives 6.41 . In both cases, OLS dramatically understates the causal effect. This is exactly the signature of the labor-withdrawal mechanism from the New Economics of Labor Migration: sending a household member abroad generates income through remittances but simultaneously reduces on-farm labor supply, decreasing food production and partially offsetting the income gain (Stark and Bloom, 1985; Lucas and Stark, 1985). The net OLS coefficient thus reflects both a positive income effect and a negative labor supply effect, and the two partially cancel. The IV, by isolating the income channel through exogenous variation in destination-country conditions, recovers the pure insurance and income effect. This interpretation is directly supported by the Hausman test statistics, which range from 4.33 to 68.81 across outcomes and reject OLS consistency for every welfare dimension in the smallholder sample. This gap carries a practical implication: studies that rely primarily on OLS, which characterizes much of the Nepal-specific literature prior to this study (Thapa and Acharya, 2017; Wagle and Devkota, 2018), likely understate the welfare benefits of remittances and may lead policymakers to under-invest in mechanisms that facilitate remittance flows.

Importantly, this empirical divergence is not a localized anomaly but a structural reality across global migration corridors. Whenever migration decisions involve a trade-off between domestic labor allocation and overseas income generation, OLS estimates will yield an incomplete, attenuated picture of household welfare. By isolating the exogenous shift-share destination channel, our identification strategy uncovers the true magnitude of the remittance income effect, offering an empirical approach that can be directly applied to evaluate migration-driven safety nets across other major remittance-dependent regions worldwide.

4.3. *The Buffering Effect: Remittances as Insurance Against Shocks*

The IV-TWFE buffering model asks not just whether remittances improve average welfare, but whether they specifically protect households when adverse shocks strike. The results provide clear affirmative evidence, with effect magnitudes that are substantively meaningful. When a shock hits, the HFIAS score increases by 5.50 points on average, a sharp increase that reflects the anxiety, dietary reduction, and deprivation experienced by smallholder households during droughts, floods, price hikes, or health crises. Among households receiving overseas remittances, this spike is dampened by 1.045 points per log-unit of remittance income ($\theta = -1.045$; $p = 0.005$), which in mean remittance amounts corresponds to absorbing approximately 19% of the total shock-induced deterioration. The FIES score

shows a parallel pattern: shocks increase it by 4.005 points, while remittances offset 0.753 of that increase ($\theta = -0.753$; $p = 0.001$). For binary food insecurity indicators, the buffering effects are even more intuitive in policy terms: households receiving remittances face a 20.2 percentage point lower probability of crossing the any-insecurity threshold following a shock ($\theta = -0.202$; $p < 0.001$), and a 10.1 percentage point lower probability of moderate-or-severe insecurity ($\theta = -0.101$; $p = 0.016$). And for the food consumption reduction indicator, the outcome that captures the most acute coping behavior, the buffering coefficient of -0.118 ($p = 0.031$) essentially cancels the direct shock effect of 0.083. Remittance-receiving households, when hit by a shock, barely resort to cutting meals at all.

This is precisely what the NELM insurance hypothesis predicts. Stark and Bloom (1985) and Lucas and Stark (1985) argued that households send members to geographically separate labor markets not just to increase income on average, but specifically to create an income stream that is uncorrelated with local shocks. Our results confirm that this mechanism operates in practice in Nepal’s smallholder sector, not just at a theoretical level. The counter-cyclical pattern we document is consistent with Yang and Choi’s (2007) classic finding that Filipino households with overseas migrants receive higher remittances when local rainfall shocks reduce farm income, and with Combes and Ebeke’s (2011) cross-country evidence that remittances stabilize household consumption during natural disasters and financial crises. More recently, Deyal (2024b) shows that remittance flows to developing countries increase sharply in the month following natural disasters, confirming the counter-cyclical mechanism at the macro level. What is new in our paper is the direct, household-level causal estimate of how much of the shock-induced welfare loss remittances actually offset—and for Nepal specifically, where the shock environment is severe and the formal insurance market is essentially non-existent for smallholder households.

The null findings of the buffering model are equally informative. The buffering coefficients for FCS, HDDS, and log food value are not statistically significant. Remittances protect households from the worst deterioration in food access during shocks, but do not appear to preserve dietary diversity and quality during these episodes. This distinction makes intuitive sense. When a household is under acute financial stress from a shock, the priority is to keep food on the table, not to maintain variety. The fact that HFIAS and FIES, which capture exactly this kind of acute insecurity, show strong buffering effects, while FCS and HDDS—which measure dietary breadth—do not, tells a coherent story about how households prioritize the deployment of remittance income under stress. This finding is consistent with Roshania and Haardörfer (2025), who show that remittances in rural India operate primarily through market purchases rather than agricultural investment, allowing households to buy food they cannot produce, rather than improving dietary diversity through production.

4.4. *The NELM Temporal Dynamic: Labor Withdrawal Then Recovery*

The PSM-DiD estimates in Table 5 add a temporal dimension to the story that the IV-2SLS estimates cannot provide. Specification A, which captures the medium-term welfare change from 2016 to the 2017–2018 average, shows no significant improvement in food insecurity indicators, and dietary quality outcomes remain largely null, even as log annual food expenditure increases significantly (ATT = +0.042; $p = 0.006$). Specification B, which

isolates only the 2017-2018 change, reveals a positive per capita FCS effect ($ATT = +0.376$; $p = 0.051$) and a sign reversal in the reduction of food consumption. Together, these two specifications tell a story consistent with NELM: in the early post-migration period, households absorb the labor-withdrawal shock, and welfare gains remain uneven; as remittances accumulate over a full migration cycle, dietary quality begins to recover. This temporal pattern helps reconcile the conflicting findings in the cross-sectional literature, where some studies report welfare improvements from remittances, and others find null or even negative effects: the discrepancy appears to depend critically on where a household stands in the migration cycle.

This PSM-DiD result is consistent with Paudel et al. (2025), who find that while migration increases off-farm income through remittances in Nepal, it reduces wheat productivity by 7.3% due to labor shortages—a direct manifestation of the labor-withdrawal cost during the early post-migration period. It also aligns with Romano et al. (2020), who find in Bangladesh that migration-induced labor losses reduce households’ own food production even as remittance income relaxes liquidity constraints. What is notable in our PSM-DiD results is that the negative food security consequences of the labor-withdrawal phase are relatively contained: the food consumption reduction indicator shows a significant negative ATT under Specification A, meaning that even in the medium run, remittance-receiving households are less likely to cut meals than comparable non-receiving households. The insurance function appears to operate even before the full income accumulation benefit arrives.

4.5. *Why Smallholders? The Structural Logic of Concentrated Effects*

The focus on smallholder households in this paper is not a convenience; it reflects a deliberate theoretical choice that the results verify. Table 1 shows that smallholder and non-smallholder households experience virtually identical shock prevalence (44.6 versus 43.8 percent; $\chi^2 = 1.00$; $p = 0.317$) and identical propensity to receive remittances (57.4 versus 58.2%; $\chi^2 = 0.60$; $p = 0.439$). However, smallholders have HFIAS scores almost twice as high (1.073 versus 0.553; $F = 146.70$; $p < 0.001$) and face a 7.8 percentage point higher probability of food insecurity. The difference is not in how much risk they face, but in how much of that risk they can absorb—a distinction that is increasingly emphasized in the recent resilience literature, which frames smallholder vulnerability as a function of limited absorptive and adaptive capacity rather than differential exposure (Shilomboleni et al., 2024). With cultivated area averaging just 0.30 hectares, smallholder households have an extremely narrow productive base: a drought that destroys a third of the crop is a minor setback for a household with a hectare of land and some off-farm income, but it can be catastrophic for a household where that crop is the only meaningful income source. Recent sub-Saharan evidence documents precisely this mechanism, showing that land holdings below one hectare drive structurally lower recovery capacity before subsequent shocks and generate cumulative food insecurity even when single-shock exposure is comparable to that of larger farmers (Chisengele and Nyanga, 2026; Kone et al., 2024).

Credit constraints compound this structural disadvantage: credit-constrained rates differ significantly between the two groups, Table 1, confirming that smallholders cannot easily borrow to smooth consumption over a bad season. Evidence from Nigeria’s LSMS-ISA

panel shows that smallholders face binding constraints on both sides: on the supply side, due to lack of collateral, and on the demand side, due to risk aversion and transaction costs—problems that are worse for farmers with smaller landholdings (Balana and Oyeyemi, 2022). Remittances fill both gaps simultaneously—they provide income that is not correlated with local agricultural conditions, and they relax the liquidity constraint that would otherwise force households into distress borrowing or asset sales. Recent evidence confirms this dual function, with remittances acting as a consumption-smoothing instrument that substitutes for missing credit markets and protects productive assets from distress liquidation (Monwanou and Akpa, 2025; Ibok et al., 2025). Nepalese evidence is particularly instructive: IV-based analysis shows that remittances significantly expand food, education, and agricultural-investment expenditures while reducing food insecurity, with effects concentrated in lower-productivity and land-constrained households (Thapa-Parajuli et al., 2025).

To verify that the smallholder restriction reflects this structural logic rather than a convenient selection of the subsample that happens to yield significant results, we re-estimate the full kernel-weighted PSM-DiD design—identical specification, covariates, kernel, and bandwidth rule—on two alternative samples: the full pooled sample of smallholder and non-smallholder households (Appendix Table A7), and non-smallholder households only (Appendix Table A8). The comparison across Tables 5, A7, and A8 supports a heterogeneity interpretation rather than a specification-search concern. In the complete pooled sample, the Reduced Food Consumption coping outcome remains significantly negative under Specification A ($ATT = -0.023$; $p = 0.007$), consistent with the smallholder-only result, but Specification B—the recovery-phase window that produces the positive dietary-quality finding among smallholders—shows no significant effect for any outcome. In the non-smallholder-only sample, the pattern weakens further: only two Specification A outcomes reach significance (Log PC Food Value and Per Capita FCS, both negative), and Specification B is again null throughout. Non-smallholder households, who by construction have larger landholdings, more diversified income sources, and—as shown above—markedly better baseline food security, appear to have alternative buffers against shocks that make overseas remittances a less pivotal insurance mechanism; the absence of a detectable recovery-phase effect in this subgroup is therefore consistent with, rather than a threat to, the paper’s central claim. We report both alternative-sample tables in full, including all non-significant estimates, so that the reader can evaluate the concentration of effects among smallholders directly rather than relying on the smallholder-only results in isolation.

5. Conclusion and Policy Implications

This paper presents the first quasi-experimental panel evidence on the causal welfare effects of overseas remittances in Nepal, including their capacity to buffer smallholder households against shock-induced welfare losses. Three complementary identification strategies converge on a consistent finding: overseas remittances significantly reduce baseline food insecurity (HFIAS: -0.825 points; FIES: -0.610 points; both $p < 0.001$) and compensate for roughly 19% shock-induced deterioration in these indicators among smallholder households. These effects are concentrated among smallholder households, reflecting tighter income bases

and binding credit constraints rather than differential shock exposure, and operate primarily through consumption smoothing rather than uniform dietary upgrading, consistent with the New Economics of Labor Migration prediction.

The most direct implication of this study is that Nepal’s formal social protection system and its private remittance economy are not substitutes—they address different parts of the same problem and both need to work better. Public cash transfer programs currently reach only about 35% of rural households and provide fixed benefits that do not expand when a drought strikes or a flood destroys a harvest. The buffering coefficients estimated in this study, $\theta = -1.045$ for HFIAS and -0.753 for FIES (both $p < 0.01$), show that remittances quietly fill this gap for households lucky enough to receive them. Rather than trying to replicate that mechanism through new expensive government programs, the most cost-effective policy response is to efficiently manage the existing channels of private capital by eliminating barriers in the system. Nepal has already reduced the average transfer costs to 3.7%, within striking distance of the SDG 10.c target of less than 3%; closing this gap would directly translate into higher net transfers for the poorest receiving households.

Beyond remittance facilitation, the PSM-DiD evidence points to a specific vulnerability that standard policies miss: the welfare gap that opens between the moment a household member leaves and the moment remittance flows fully materialize. The 2016-baseline PSM-DiD specification (Specification A) shows no significant improvement in food insecurity during the medium-run window, while the 2017-baseline specification (Specification B) reveals dietary recovery in the terminal year—a temporal sequence that precisely mirrors the labor-withdrawal and remittance-accumulation dynamic predicted by the New Economics of Labor Migration.

This transitional gap is where smallholder households are most exposed and where intervention would pay off most. No single program can close it; global experience points to a sequenced set of instruments instead. Seasonal public employment guarantees, on the model of India’s MGNREGA and similar sub-Saharan African schemes, can bridge the welfare dip at planting and harvest, while conditional cash transfers—as in the Philippines’ 4Ps, Indonesia’s Family Hope Program and Nigeria’s Household Upliftment Program—provide a consumption floor until remittances arrive. These transitional measures work best paired with reforms that increase migration’s long-term value: digitizing remittance corridors toward the G20’s three-percent cost target, bundling weather-index insurance with credit and inputs as piloted in Kenya, Ethiopia, and India, and extending smallholder credit guarantees on Tanzania’s model to help households convert liquidity into investment. A unified beneficiary registry, modeled on India’s eShram platform and the IFAD’s programs in Tajikistan and Uzbekistan, would reduce administrative friction and accelerate delivery to households that need it the most.

The heterogeneity evidence reinforces the case for this layered approach: because the food insecurity gap between smallholders and non-smallholders reflects lower coping capacity rather than higher shock exposure, universal agricultural support programs will always under-perform relative to instruments calibrated to the structural constraints of land-constrained households. Closing the coping capacity gap, rather than the shock exposure gap, is the right policy objective.

6. Limitations of this study

Although the findings offer robust and consistent evidence, several limitations should be noted. First, our instrumental variable strategy identifies a Local Average Treatment Effect (LATE) specific to "complier" households—those whose remittance inflows are sensitive to destination-country macroeconomic conditions. The extent to which these patterns generalize to households driven by different migration dynamics remains an open empirical question. Future work leveraging alternative instruments, such as bilateral exchange rate shocks or shifting migration policies, would help benchmark this external validity.

Second, our three-year panel captures only a segment of the New Economics of Labor Migration (NELM) welfare trajectory. Although the PSM-DiD estimates suggest a continuing post-shock recovery beyond Wave 3, a longer panel is required to definitively trace long-run income and asset accumulation mechanisms. Third, because we treat remittance income as a pooled flow, we cannot disentangle the distinct channels, such as consumption smoothing, debt retirement, or human capital investment, driving the observed insurance effects. The disaggregation of these flows represents a fertile area for future study.

Finally, the gender dimension of migration-induced household restructuring, visible in the 19.3 percentage-point female-headship gap between smallholder and non-smallholder households, raises questions about how the welfare effects are distributed within households when migration leaves women managing small farms alone. These are important questions that current data cannot fully resolve and that future research should address directly.

Data availability

Household data are from the Nepal Household Risk and Vulnerability Survey (HRVS), waves 2016–2018, collected by the World Bank in collaboration with the Government of Nepal. The data is third-party and not owned by the authors. Interested researchers can access the data set through the World Bank Microdata Library by registering and submitting a data request in the same manner as the authors (Study ID: NPL_2016-2018_HRVS_v01_M; <https://microdata.worldbank.org/>), subject to the standard terms of use of the World Bank. There are no restrictions beyond registration and request. Destination-country GDP growth data are publicly available without restriction from the World Bank's World Development Indicators (Indicator: NY.GDP.MKTP.KD.ZG; <https://data.worldbank.org/>).

Appendix

Table A1: Descriptive Statistics by Survey Wave, Smallholder Households (≤ 0.5 ha), Nepal HRVS 2016–2018

Variable	N	2016	2017	2018	Min	Max
Panel A: Welfare Outcomes						
Log PC Food Value (7-day)	12,189	5.908 (0.464)	5.892 (0.495)	5.998 (0.413)	3.970	8.374
Log Annual Food Expenditure	12,686	11.377 (0.498)	11.381 (0.504)	11.443 (0.457)	9.219	13.568
Food Consumption Score (FCS)	12,688	78.877 (24.650)	80.585 (22.769)	84.033 (21.615)	0.000	112.000
Per Capita FCS	12,190	20.651 (12.133)	21.297 (12.254)	22.733 (12.976)	0.000	112.000
HFIAS Score (0–27)	12,688	1.759 (3.447)	1.120 (2.997)	0.328 (1.620)	0.000	27.000
FIES Score (0–8)	12,688	1.318 (2.340)	0.787 (1.952)	0.251 (1.124)	0.000	8.000
HDDS (0–12)	12,688	8.114 (1.645)	8.212 (1.495)	8.562 (1.431)	0.000	12.000
Any Food Insecure – HFIAS (0/1)	12,688	0.283 (0.451)	0.163 (0.369)	0.056 (0.230)	0.000	1.000
Mod/Sev Food Insecure – FIES (0/1)	12,688	0.216 (0.412)	0.129 (0.335)	0.045 (0.208)	0.000	1.000
Reduced Food Consumption (0/1)	12,921	0.069 (0.253)	0.036 (0.186)	0.015 (0.121)	0.000	1.000
Panel B: Remittances						
Log Overseas Remittance Amount (NRs)	4,534	8.164 (4.754)	10.849 (1.647)	10.773 (1.678)	0.000	15.493
Received Overseas Remittance (0/1)	7,255	0.285 (0.451)	0.989 (0.105)	0.993 (0.084)	0.000	1.000
Panel C: Shocks						
Experienced Any Shock (0/1)	12,921	0.792 (0.406)	0.327 (0.469)	0.222 (0.416)	0.000	1.000
Natural Shock (0/1)	12,921	0.589 (0.492)	0.145 (0.352)	0.106 (0.308)	0.000	1.000
Economic Shock (0/1)	12,921	0.211 (0.408)	0.005 (0.069)	0.003 (0.051)	0.000	1.000
Panel D: Household Controls						
Age of HH Head (years)	12,333	47.357 (13.818)	47.997 (14.014)	48.811 (13.921)	15.000	98.000
Household Size	12,184	4.721 (1.852)	4.531 (1.860)	4.324 (1.822)	1.000	16.000
Years of Education (HH Head)	12,209	3.709 (4.248)	3.692 (4.278)	3.675 (4.269)	0.000	18.000
Panel E: Instrument						
Bartik IV (Z_{dt})	12,688	1.638 (0.636)	1.409 (0.697)	1.115 (0.614)	0.313	3.117

Notes: Sample restricted to smallholder households (cultivated area ≤ 0.5 ha). N is the total number of non-missing household-wave observations pooled across all three waves; Min and Max are computed on the same pooled sample. HFIAS = Household Food Insecurity Access Scale; FIES = Food Insecurity Experience Scale; HDDS = Household Dietary Diversity Score; FCS = Food Consumption Score; HH = Household. The Bartik instrument (Z_{dt}) is a shift-share index of district-level remittance exposure constructed from pre-2016 migrant destination shares interacted with lagged destination-country GDP growth rates.

Table A2: Hausman Endogeneity Tests – All Outcomes and Subsamples

Welfare Outcome	R1 –SM		R2 –SM		R1 –NS		R2 –NS		R2 –Full	
	H-stat	Endog.?	H-stat	Endog.?	H-stat	Endog.?	H-stat	Endog.?	H-stat	Endog.?
Log PC Food Value (7-day)	30.91	Yes	19.93	Yes	29.07	Yes	23.84	Yes	53.18	Yes
Log Annual Food Expenditure	24.66	Yes	11.11	Yes	30.49	Yes	21.33	Yes	38.15	Yes
Food Consumption Score (FCS)	32.51	Yes	25.89	Yes	14.11	Yes	10.15	Yes	39.68	Yes
Per Capita FCS	4.33	Yes	4.50	Yes	3.71	No	1.82	No	7.56	Yes
HFIAS Score	23.52	Yes	29.08	Yes	0.66	No	1.56	No	37.70	Yes
FIES Score (0–8)	28.78	Yes	39.37	Yes	0.55	No	1.35	No	47.89	Yes
HDDS Score (0–12)	68.81	Yes	63.53	Yes	23.23	Yes	16.48	Yes	86.87	Yes
Any Food Insecure – HFIAS (0/1)	64.73	Yes	66.94	Yes	3.32	No	5.08	Yes	85.63	Yes
Mod/Sev Food Insecure – FIES (0/1)	20.62	Yes	35.52	Yes	0.34	No	1.62	No	41.99	Yes
Reduced Food Consumption (0/1)	46.91	Yes	32.24	Yes	5.46	Yes	4.25	Yes	35.94	Yes
$\chi^2(1)$ critical value (5%)	3.84	—	3.84	—	3.84	—	3.84	—	3.84	—

Notes: H-stat = Hausman test statistic $\chi^2(1)$. H_0 : OLS consistent (R_{it} exogenous). “Yes” indicates rejection at the 5% level (H -stat > 3.84). SM = Smallholder; NS = Non-Smallholder. “Mixed” for NS reflects weaker instrument strength in the NS subsample.

Table A3: First-Stage Regressions–IV Relevance Check

Welfare Outcome	R1–SM, Direct			R1–SM, Buffering			R2–NS, Direct			R2–NS, Buffering		
	KP F -stat	N	Instrument(s)	KP F -stat	N	Instrument(s)	KP F -stat	N	Instrument(s)	KP F -stat	N	Instrument(s)
Log PC Food Value (7-day)	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
Log Annual Food Expenditure	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
Food Consumption Score (FCS)	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
Per Capita FCS	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
HFIAS Score	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
FIES Score (0–8)	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
HDDS Score (0–12)	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
Any Food Insecure – HFIAS (0/1)	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
Mod/Sev Food Insecure – FIES (0/1)	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
Reduced Food Consumption (0/1)	63.0	6,073	Z_{dt}	27.3	6,073	$Z_{dt}, Z_{dt} \times S_{it}$	47.4	4,209	Z_{dt}	16.7	4,209	$Z_{dt}, Z_{dt} \times S_{it}$
Stock–Yogo CV (10%)	16.38			16.38			16.38			16.38		
Passes weak-IV test?	Yes (63.0 > 16.38)			Yes (27.3 > 16.38)			Yes (47.4 > 16.38)			Yes (16.7 > 16.38)		

Notes: First-stage model: $R_{it} = \pi_0 + \pi_1 Z_{dt} + \pi_2 S_{it} + X'\theta + \text{FE} + \nu$ (FS1, direct model); adds $Z_{dt} \times S_{it}$ for FS2 (buffering model). KP F -stat uniform across outcomes for given model. The buffering model with R2 (FS2, KP- $F = 16.7$) falls slightly above the Stock–Yogo threshold; Fixed effects & controls: All models include HH fixed effects and survey-year fixed effects. Time-varying household controls: age and gender of head, household size, years of education, ethnicity, debt status, savings status, and public transfer receipt. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Reduced Form Estimates — Bartik IV (Z_{dt}) Direct Effect on Welfare

Welfare Outcome	R1 IV — Smallholder			R2 IV — Smallholder			R2 IV — Full Sample		
	RF Coef.	p -val	Sig.	RF Coef.	p -val	Sig.	RF Coef.	p -val	Sig.
Log PC Food Value (7-day)	0.0435	< 0.001	***	0.0466	< 0.001	***	0.0622	< 0.001	***
Log Annual Food Expenditure	0.0379	< 0.001	***	0.0367	< 0.001	***	0.0528	< 0.001	***
Food Consumption Score (FCS)	2.4632	0.047	**	3.0597	< 0.001	***	3.0459	< 0.001	***
Per Capita FCS	0.3509	0.076	*	0.5516	0.025	**	0.5588	0.005	***
HFIAS Score	-0.2710	< 0.001	***	-0.3938	< 0.001	***	-0.3542	< 0.001	***
FIES Score (0–8)	-0.1962	< 0.001	***	-0.2911	< 0.001	***	-0.2583	< 0.001	***
HDDS Score (0–12)	0.2439	0.081	*	0.3102	< 0.001	***	0.2930	< 0.001	***
Any Food Insecure – HFIAS (0/1)	-0.0610	< 0.001	***	-0.0760	< 0.001	***	-0.0699	< 0.001	***
Mod/Sev Food Insecure – FIES (0/1)	-0.0297	< 0.001	***	-0.0506	< 0.001	***	-0.0443	< 0.001	***
Reduced Food Consumption (0/1)	-0.0365	< 0.001	***	-0.0304	< 0.001	***	-0.0281	< 0.001	***
N (median)	$\approx 6,073$			$\approx 4,209$			$\approx 6,080$		
HH fixed effects	Yes			Yes			Yes		
Year fixed effects	Yes			Yes			Yes		
Within R^2	0.09–0.34			0.06–0.22			0.09–0.38		

Notes: Reduced-form estimates: $W_{it} = \alpha + \pi RFZ_{dt} + \beta S_{it} + X'_{it}\delta + FE + e$. RF Coef. = intent-to-treat effect of Bartik IV on welfare. The 2SLS estimate recovers $\pi/\hat{\pi}_1$ (Wald identity). Within R^2 varies by outcome. Fixed effects and controls: all models include household fixed effects and survey-year fixed effects. Time-varying household controls are age and gender of household head, household size, years of education, ethnicity, debt status, savings status, and public transfer receipt. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A5: PSM-DiD Bandwidth Sensitivity–Smallholder Households Only

ATT estimated at multiples of the Silverman optimal bandwidth. Stability of signs across all multipliers confirms robustness of main PSM-DiD results.

Welfare Outcome	$\times 0.50$		$\times 0.75$		Ref. ($\times 1.00$)		$\times 1.25$		$\times 1.50$	
	ATT	Sig.	ATT	Sig.	ATT	Sig.	ATT	Sig.	ATT	Sig.
<i>Panel A: Specification A — 2016 Baseline (Reference BW = 0.031, N = 4,287)</i>										
Log PC Food Value (7-day)	0.015	ns	0.014	ns	0.015	ns	0.015	ns	0.015	ns
Log Annual Food Expenditure	0.042	***	0.042	***	0.042	***	0.042	***	0.042	***
Food Consumption Score (FCS)	0.226	ns	0.152	ns	0.116	ns	0.098	ns	0.080	ns
Per Capita FCS	1.349	***	1.348	***	1.353	***	1.370	***	1.380	***
HFIAS Score	−0.628	***	−0.631	***	−0.627	***	−0.624	***	−0.621	***
FIES Score (0–8)	−0.387	***	−0.387	***	−0.384	***	−0.381	***	−0.378	***
HHDS Score (0–12)	0.009	ns	0.008	ns	0.008	ns	0.008	ns	0.008	ns
Any Food Insecure – HFIAS (0/1)	−0.067	***	−0.066	***	−0.065	***	−0.065	***	−0.065	***
Mod/Sev Insecure – FIES (0/1)	−0.069	***	−0.069	***	−0.068	***	−0.067	***	−0.067	***
Reduced Food Consumption (0/1)	−0.020	*	−0.019	*	−0.018	*	−0.019	*	−0.019	*
<i>Panel B: Specification B — 2017 Baseline (Reference BW = 0.038, N = 4,358)</i>										
Log PC Food Value (7-day)	0.026	ns	0.026	ns	0.026	ns	0.026	ns	0.027	ns
Log Annual Food Expenditure	0.001	ns	0.001	ns	0.002	ns	0.002	ns	0.003	ns
Food Consumption Score (FCS)	0.524	ns	0.507	ns	0.540	ns	0.546	ns	0.538	ns
Per Capita FCS	0.386	**	0.370	*	0.376	*	0.382	**	0.388	**
HFIAS Score	−0.046	ns	−0.050	ns	−0.050	ns	−0.051	ns	−0.054	ns
FIES Score (0–8)	−0.038	ns	−0.038	ns	−0.037	ns	−0.038	ns	−0.040	ns
HHDS Score (0–12)	0.020	ns	0.020	ns	0.021	ns	0.021	ns	0.021	ns
Any Food Insecure – HFIAS (0/1)	−0.014	ns	−0.014	ns	−0.014	ns	−0.014	ns	−0.015	ns
Mod/Sev Insecure – FIES (0/1)	−0.008	ns	−0.008	ns	−0.008	ns	−0.008	ns	−0.008	ns
Reduced Food Consumption (0/1)	0.002	ns	0.002	ns	0.002	ns	0.001	ns	0.001	ns

Notes: ATT = Average Treatment Effect on the Treated. Bandwidth multipliers: $\times \{0.50, 0.75, 1.00$ (Silverman reference), $1.25, 1.50\}$. Epanechnikov kernel; Abadie–Imbens (2006) standard errors. Both specifications: smallholder stratum (≤ 0.5 ha), HH and year fixed effects, all household controls. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; ns = not significant.

Table A6: Propensity Score Covariate Balance–Before Matching (Both Specifications)

Design element	Specification A (2016 Baseline)	Specification B (2017 Baseline)
Sample	Smallholder HHs (≤ 0.5 ha)	Smallholder HHs (≤ 0.5 ha)
Treatment	Received overseas remittance (0/1)	Received overseas remittance (0/1)
Pre-period	Wave 2016	Wave 2017
Post-period	Mean of Waves 2017 & 2018	Wave 2018
PS Model	Logistic (L2, $C = 1.0$), standardised	Logistic (L2, $C = 1.0$), standardised
AUC	0.717	0.749
Silverman BW	0.031	0.038
Common support	[0.043, 0.751]	[0.086, 0.824]
N treated	1,221	1,409
N control	3,066	2,949

Panel A: Specification A – 2016 Baseline ($N_{treat} = 1,221$; $N_{ctrl} = 3,066$; $BW = 0.031$; $AUC = 0.717$)

Covariate	Treated	Control	SMD	SMD <0.10?	Balance
Age of HH Head (years)	47.990	46.951	0.077	Yes	Well balanced
Household Size	4.591	4.804	−0.117	No	Marginal
Years of Education (HH Head)	2.848	3.259	−0.155	No	Marginal
Has Outstanding Debt (0/1)	0.478	0.430	0.098	Yes	Well balanced
Has Any Savings (0/1)	0.869	0.784	0.120	No	Marginal
Receives Public Transfer (0/1)	0.346	0.378	−0.065	Yes	Well balanced
Owns Non-Agri Enterprise (0/1)	0.089	0.133	−0.142	No	Marginal

Panel B: Specification B – 2017 Baseline ($N_{treat} = 1,409$; $N_{ctrl} = 2,949$; $BW = 0.038$; $AUC = 0.749$)

Covariate	Treated	Control	SMD	SMD <0.10?	Balance
Age of HH Head (years)	48.215	47.836	0.027	Yes	Well balanced
Household Size	4.359	4.584	−0.122	No	Marginal
Years of Education (HH Head)	3.014	3.545	−0.095	Yes	Well balanced
Has Outstanding Debt (0/1)	0.661	0.545	0.238	No	Marginal
Has Any Savings (0/1)	0.983	0.944	0.144	No	Marginal
Receives Public Transfer (0/1)	0.296	0.337	−0.088	Yes	Well balanced
Owns Non-Agri Enterprise (0/1)	0.123	0.187	−0.177	No	Marginal

Notes: SMD = standardised mean difference = $(\text{mean}_{\text{treated}} - \text{mean}_{\text{control}}) / \text{SD}_{\text{pooled}}$. $|SMD| < 0.10$ indicates a well-balanced covariate. Pre-matching imbalances in household size, education, savings, and enterprise ownership are addressed by Epanechnikov kernel weighting in the PSM-DiD estimation. AUC = area under the ROC curve of the logistic propensity score model. L2 regularisation ($C = 1.0$) applied to all logistic models.

Table A7: Robustness Check: Kernel-Weighted PSM-DiD – Full Sample (Smallholders and Non-Smallholders)

Welfare Outcome	Specification A — 2016 Baseline						Specification B — 2017 Baseline					
	ATT	SE	t-stat	p-val	CI _{lo}	CI _{hi}	ATT	SE	t-stat	p-val	CI _{lo}	CI _{hi}
Log PC Food Value (7-day)	−0.031**	(0.013)	−2.365	0.018	−0.057	−0.005	−0.023*	(0.014)	−1.676	0.094	−0.051	0.004
Log Annual Food Expenditure	0.020	(0.013)	1.463	0.144	−0.007	0.046	−0.003	(0.014)	−0.238	0.812	−0.030	0.024
Food Consumption Score (FCS)	−0.618	(0.739)	−0.837	0.403	−2.066	0.830	0.034	(0.762)	0.045	0.964	−1.460	1.528
Per Capita FCS	—	—	—	—	—	—	−0.318	(0.296)	−1.074	0.283	−0.898	0.262
HFIAS Score	—	—	—	—	—	—	−0.054	(0.078)	−0.696	0.487	−0.206	0.098
FIES Score (0–8)	—	—	—	—	—	—	−0.030	(0.051)	−0.590	0.555	−0.129	0.069
HDDS Score (0–12)	−0.050	(0.045)	−1.111	0.267	−0.137	0.038	0.042	(0.048)	0.889	0.374	−0.051	0.136
Any Food Insecure – HFIAS (0/1)	—	—	—	—	—	—	−0.010	(0.010)	−1.023	0.306	−0.031	0.010
Mod/Sev Insecure – FIES (0/1)	—	—	—	—	—	—	−0.006	(0.009)	−0.639	0.523	−0.023	0.012
Reduced Food Consumption (0/1)	−0.023***	(0.009)	−2.694	0.007	−0.040	−0.006	−0.003	(0.006)	−0.500	0.617	−0.016	0.009
<i>N</i> (treated / control)	1,721 / 3,927						1,930 / 3,899					
Kernel / Bandwidth	Epanechnikov / 0.030						Epanechnikov / 0.030					
AUC (PS model)	0.716						0.711					
Common Support	[0.075, 0.779]						[0.078, 0.809]					
HH Fixed Effects	Yes						Yes					
Year Fixed Effects	Yes						Yes					

Notes: ATT = Average Treatment Effect on the Treated (Epanechnikov kernel-weighted, Silverman bandwidth), estimated with the identical procedure as Table 5, applied to an alternative sample for robustness. Sample: all households, smallholders and non-smallholders pooled ($N = 6,000$ households, Wave 2016). Treatment: household received overseas remittance (baseline-wave status). Control: household did not. DiD outcome: Spec. A = $\bar{W}_{\text{post}} - W_{2016}$ (medium-run); Spec. B = $W_{2018} - W_{2017}$ (recovery phase). Propensity scores estimated by L2-regularised logistic regression on household controls: age and gender of head, household size, years of education, ethnicity, debt status, savings status, and public transfer receipt, and smallholder status. Standard errors from Abadie–Imbens (2006); 95% CIs reported. Dashes (—) = outcome excluded from Specification A, consistent with the convention in Table 5 of the main analysis. All values rounded to three decimal places. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8: Robustness Check: Kernel-Weighted PSM-DiD – Non-Smallholder Households Only

Welfare Outcome	Specification A — 2016 Baseline						Specification B — 2017 Baseline					
	ATT	SE	t-stat	p-val	CI _{lo}	CI _{hi}	ATT	SE	t-stat	p-val	CI _{lo}	CI _{hi}
Log PC Food Value (7-day)	−0.079***	(0.025)	−3.164	0.002	−0.127	−0.030	−0.009	(0.026)	−0.327	0.744	−0.060	0.043
Log Annual Food Expenditure	−0.043*	(0.026)	−1.661	0.097	−0.093	0.008	0.006	(0.025)	0.226	0.821	−0.044	0.055
Food Consumption Score (FCS)	−1.111	(1.400)	−0.794	0.427	−3.855	1.632	0.655	(1.405)	0.467	0.641	−2.098	3.408
Per Capita FCS	—	—	—	—	—	—	0.289	(0.569)	0.507	0.612	−0.827	1.404
HFIAS Score	—	—	—	—	—	—	−0.069	(0.137)	−0.503	0.615	−0.338	0.200
FIES Score (0–8)	—	—	—	—	—	—	−0.016	(0.084)	−0.192	0.847	−0.181	0.149
HDDS Score (0–12)	−0.099	(0.082)	−1.206	0.228	−0.260	0.062	0.121	(0.085)	1.416	0.157	−0.046	0.288
Any Food Insecure – HFIAS (0/1)	—	—	—	—	—	—	−0.003	(0.016)	−0.202	0.840	−0.035	0.029
Mod/Sev Insecure – FIES (0/1)	—	—	—	—	—	—	−0.006	(0.015)	−0.380	0.704	−0.036	0.024
Reduced Food Consumption (0/1)	−0.013	(0.014)	−0.875	0.382	−0.041	0.016	−0.020	(0.013)	−1.555	0.120	−0.046	0.005
<i>N</i> (treated / control)	496 / 1,188						539 / 1,174					
Kernel / Bandwidth	Epanechnikov / 0.026						Epanechnikov / 0.028					
AUC (PS model)	0.663						0.659					
Common Support	[0.101, 0.657]						[0.120, 0.695]					
HH Fixed Effects	Yes						Yes					
Year Fixed Effects	Yes						Yes					

Notes: ATT = Average Treatment Effect on the Treated (Epanechnikov kernel-weighted, Silverman bandwidth), estimated with the identical procedure as Table 5, applied to an alternative sample for robustness. Sample: non-smallholder households only (> 0.5 ha, $N = 1,684$ households, Wave 2016). Treatment: household received overseas remittance (baseline-wave status). Control: household did not. DiD outcome: Spec. A = $\bar{W}_{\text{post}} - W_{2016}$ (medium-run); Spec. B = $W_{2018} - W_{2017}$ (recovery phase). Propensity scores estimated by L2-regularised logistic regression on household controls: age and gender of head, household size, years of education, ethnicity, debt status, savings status, and public transfer receipt. Standard errors from Abadie–Imbens (2006); 95% CIs reported. Dashes (—) = outcome excluded from Specification A, consistent with the convention in Table 5 of the main analysis. All values rounded to three decimal places. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author did not use generative AI or AI-assisted technologies to generate, draft, analyze, or create any original content, data, results, or intellectual contributions of this manuscript. Generative AI (Claude, Anthropic) was used solely as a language-support tool to check grammar and improve the sentence structure and readability of author-written text. After using this tool, the author reviewed and edited the content as needed and took full responsibility for the content of this publication.

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